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SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Emissions, Chemical Transformation, and Deposition of
Atmospheric Reactive Nitrogen and its impact on PM_{2.5} in the
Pearl River Delta

PhD Candidate: Hanrui Duan

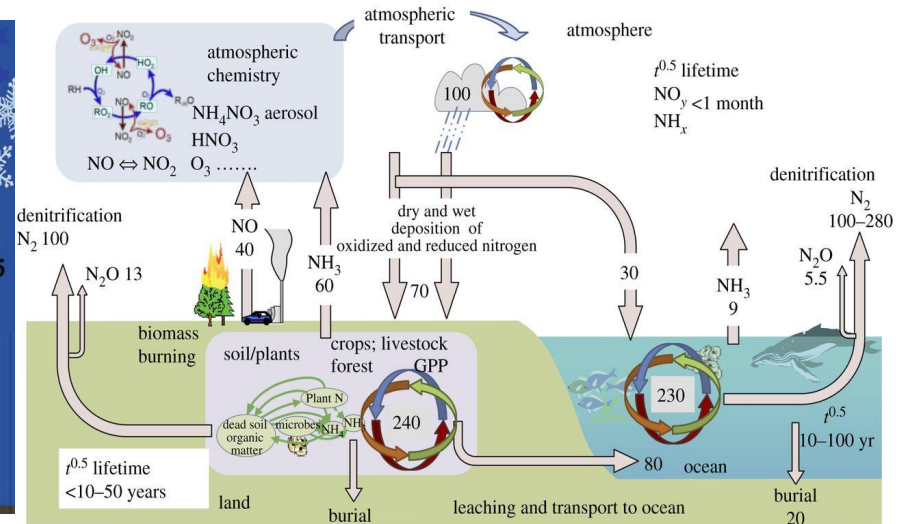
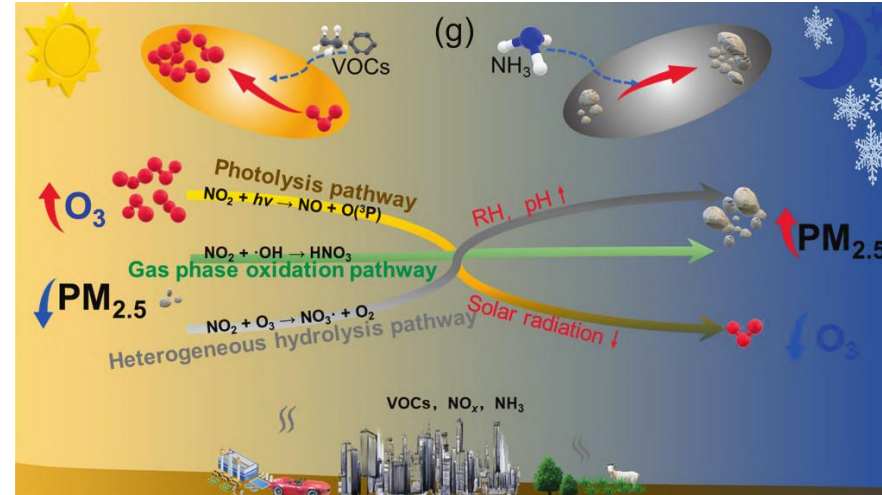
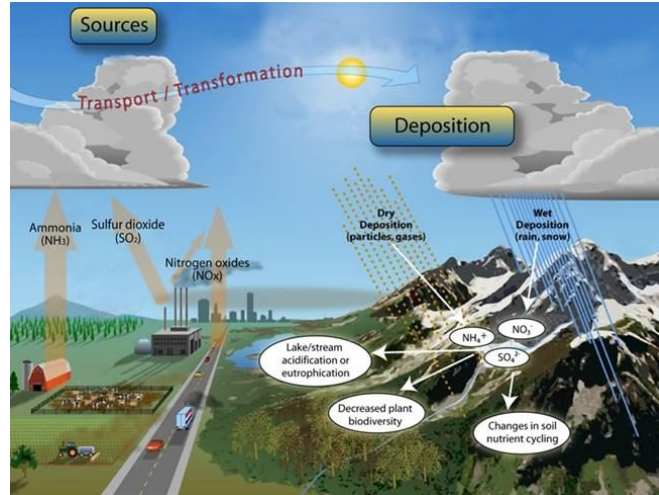
Supervisor:

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Outline

1. Background & Scientific Significance
2. Literature Review & Research Progress
3. Research Gaps & Motivation
4. Research Objectives
5. Data and Methodology
6. Preliminary Results
7. Future Work & Expected Outcomes

1. Background & Scientific Significance- Why Reactive Nitrogen Emissions Matter

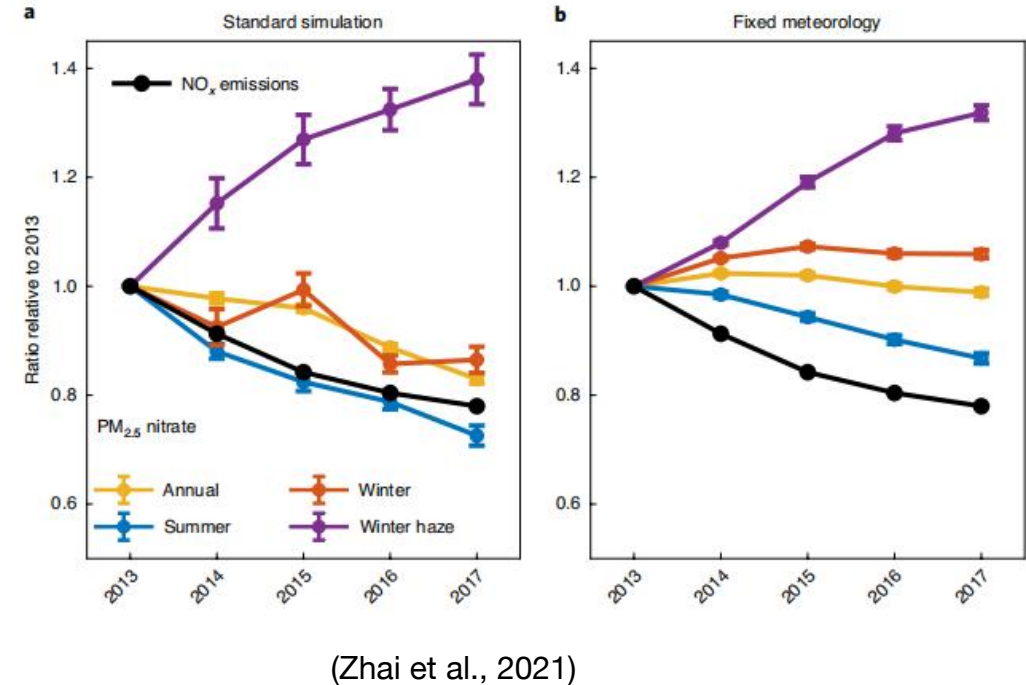
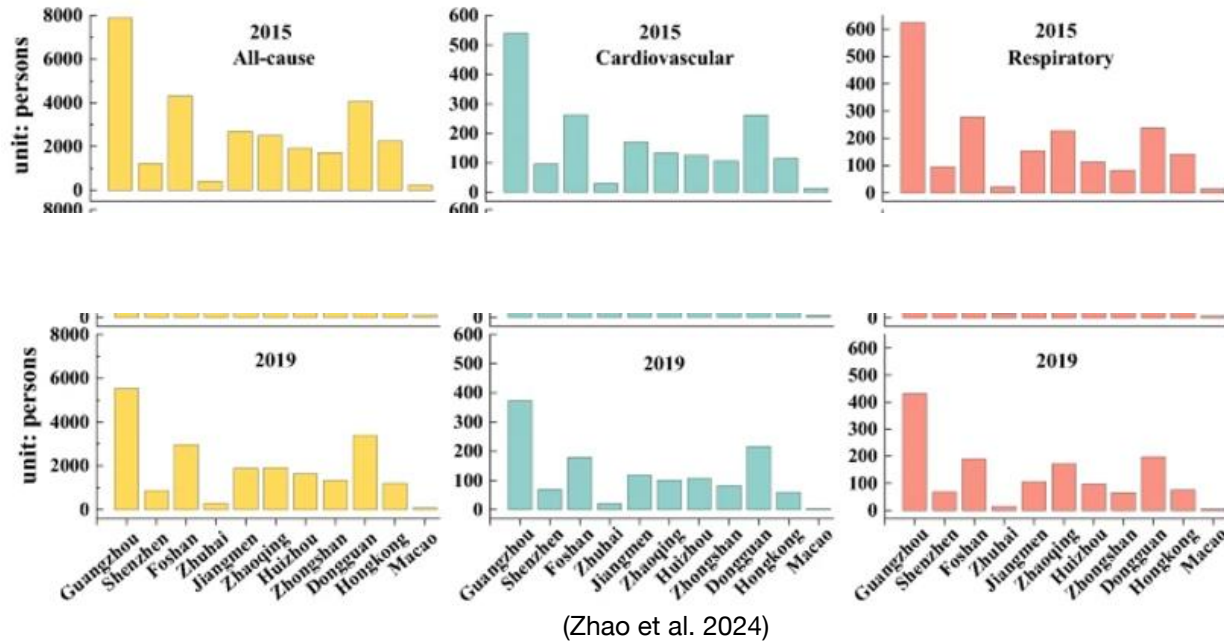


Adapted from Wang, F., et al. 2022)

Adapted from Fowler et al. (2013)

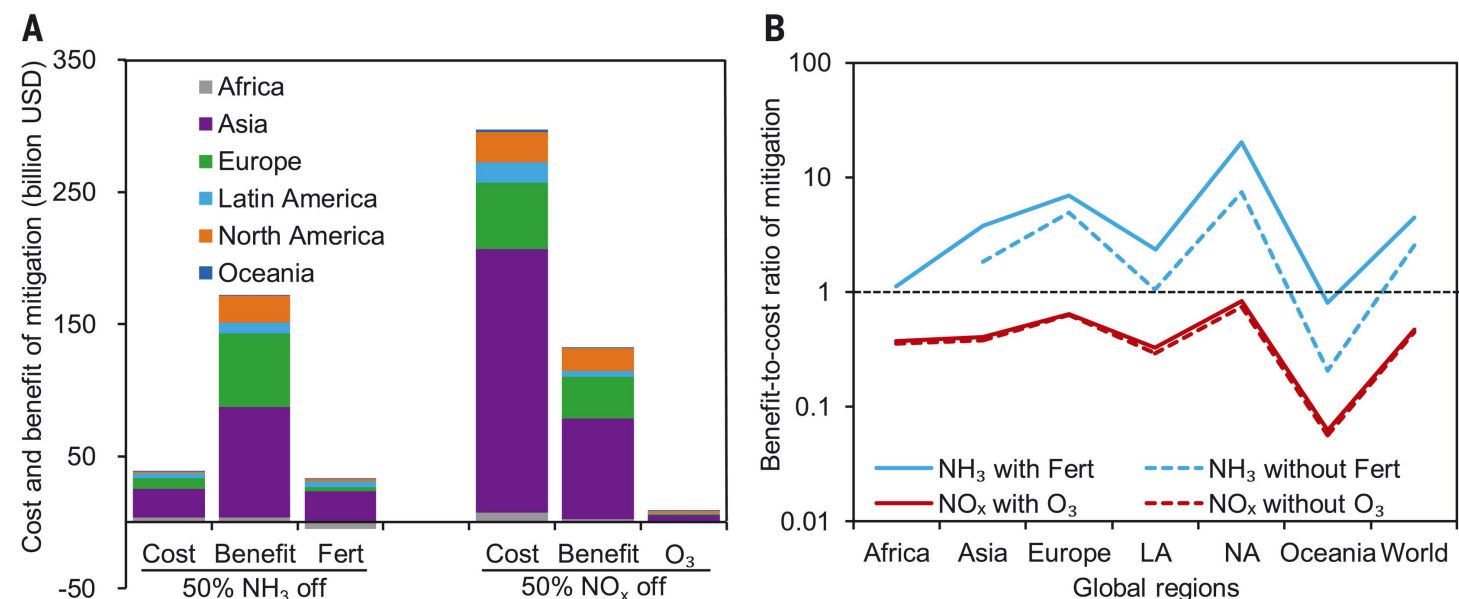
1. Reactive nitrogen (Nr) perturbations now profoundly affect air quality, climate systems, human health, and ecosystem integrity.
2. Alterations to the 21st-century global nitrogen cycle constitute one of the grandest Earth-system challenges, second only to climate change.
3. Elucidating the emissions, chemical transformation, and deposition of atmospheric Nr—and its feedback on air quality—represents a central frontier in atmospheric chemistry.

2. Literature Review 1: Nitrate Replacing Sulfate as the Dominant PM_{2.5} Component

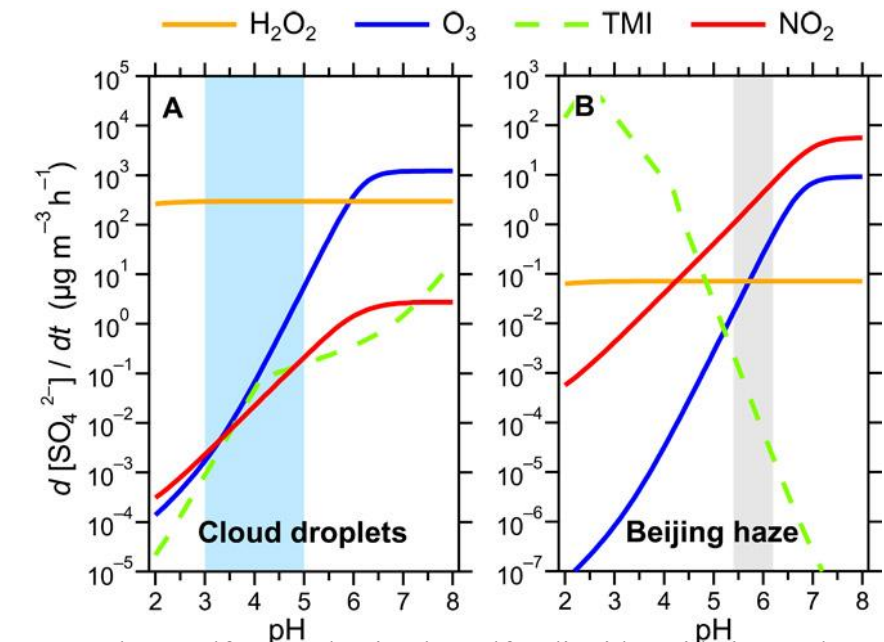


- Exposure to PM_{2.5} lead to premature deaths, cardiovascular and respiratory diseases, decreasing in PM_{2.5} concentrations can reduce these diseases.
- As the world's largest Nr emitter, China has undergone a two-decade pollution evolution: from SO₂-dominated to NO_x-NH₃-dominated regimes.
- Following substantial SO₂ abatement, particulate nitrate has become the dominant inorganic component of winter haze.

2. Literature Review 2: Rising Ammonia as a Critical Factor for PM_{2.5} Control



Cost and benefit to reduce 50% of NH₃ and NO_x emission in 2013 (Adapted from Gu et al. 2021, Fig. 3).

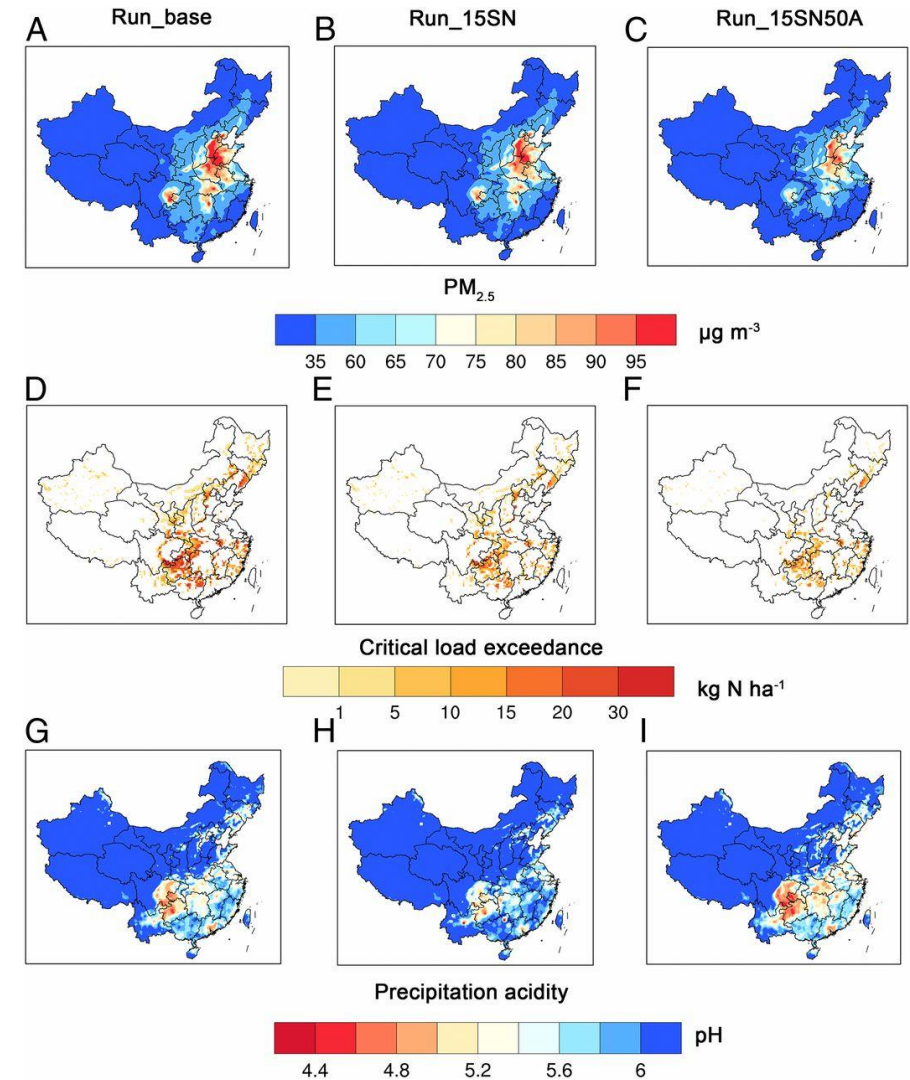


Aqueous-phase sulfate production by sulfur dioxide oxidation under characteristic conditions (Adapted from Cheng et al., 2016, Fig.2)

- NH₃ is the only abundant alkaline gas in the atmosphere. It reacts with H₂SO₄ and HNO₃ to form (NH₄)₂SO₄ and NH₄NO₃, constituting major PM_{2.5} components.
- SO₂ mitigation reduces sulfate, leaving surplus NH₃ available to combine with HNO₃ and forcing nitrate to become the dominant PM_{2.5} species.
- Globally, NH₃ remains largely unregulated. In China, NO_x fell ~38% during 2013–2020, yet NH₃ emissions stayed nearly flat—creating a blind spot for the next stage of PM_{2.5} governance.
- The mechanism by which NH₃ affects O₃ production remains an unresolved international frontier.

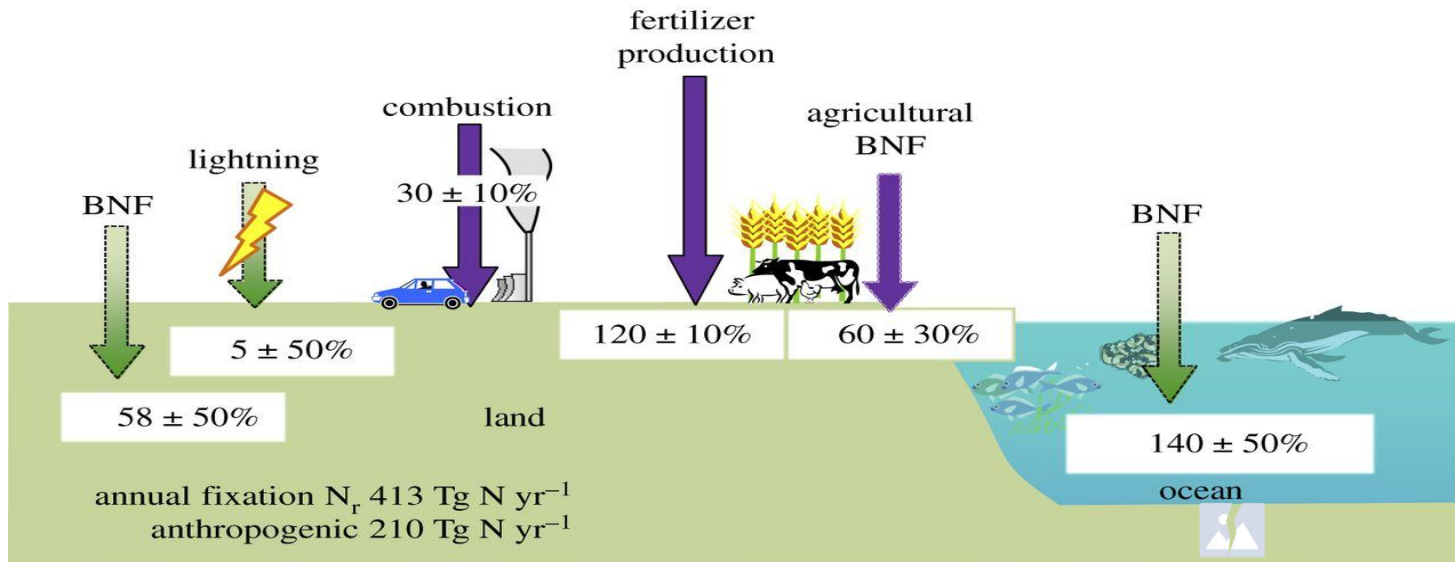
2. Literature Review 3: From NO_x -Centered to NO_x - NH_3 Coupled Research

- Early research treated NO_x as the sole lever. Over the past decade, it has been recognized that coupling oxidized nitrogen (NO_y) and reduced nitrogen (NH_x) is essential for effective $\text{PM}_{2.5}$ control.
- Changing NH_3 alters acid–base balance and nitrogen deposition. In NH_3 –rich regions, cutting NH_3 yields greater marginal $\text{PM}_{2.5}$ reductions than cutting NO_x alone.

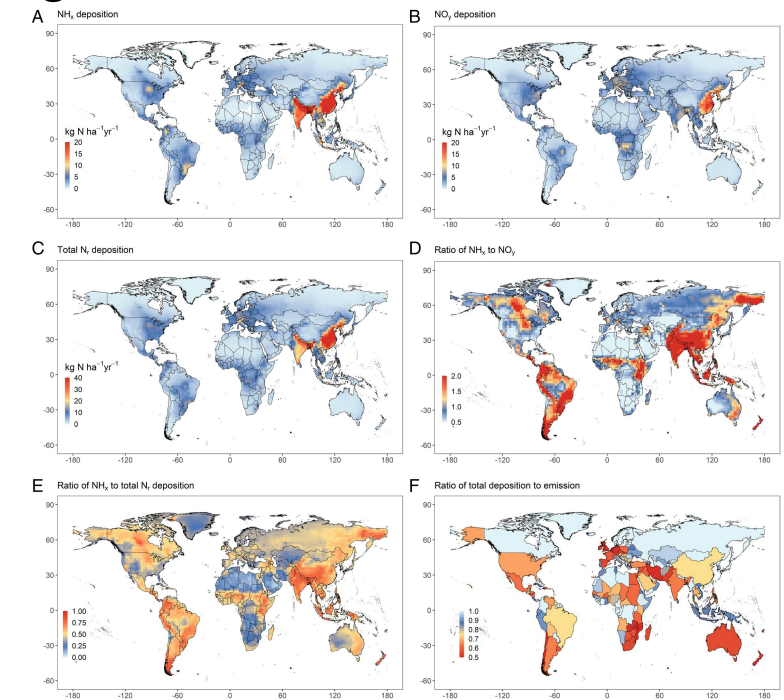


Comparison of simulated $\text{PM}_{2.5}$ concentrations (A–C), critical load exceedance of nitrogen (D–F), and precipitation-weighted mean pH (G–I) over China in the Run_base, Run_15SN, and Run_15SN50A scenarios. The small islands in the South China Sea were excluded (Adapted from Liu et al., 2019)

2. Literature Review 4: Mutual Feedback Between Reactive Nitrogen and Climate



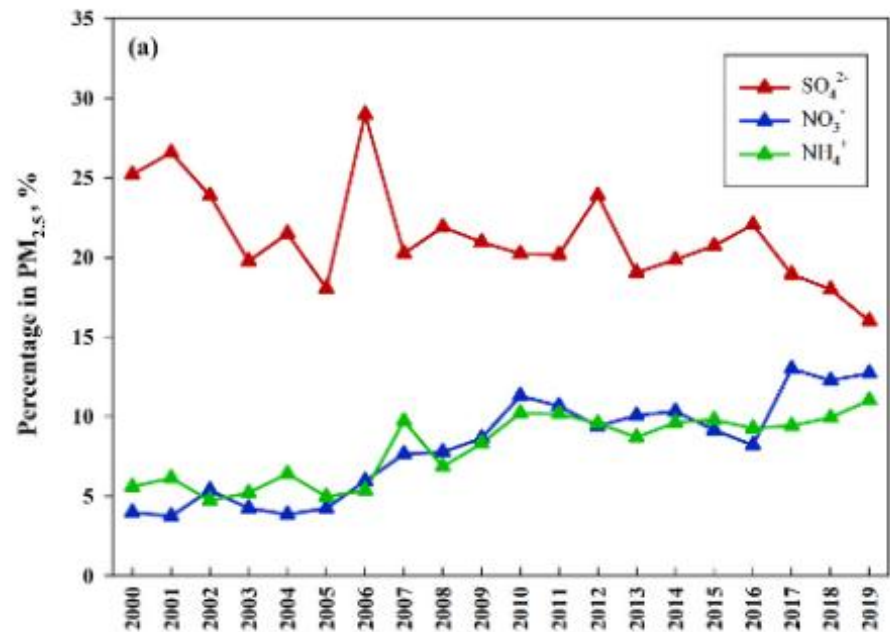
Global nitrogen fixation, natural and anthropogenic in both oxidized and reduced forms through combustion, biological fixation, lightning and fertilizer and industrial production through the Haber–Bosch process for 2010 (Adapted from Fowler et al., 2013 Figure 1)



Atmospheric N_r deposition simulated by the GEOS-Chem model in 2010 (Liu et al., 2022, Figure 3)

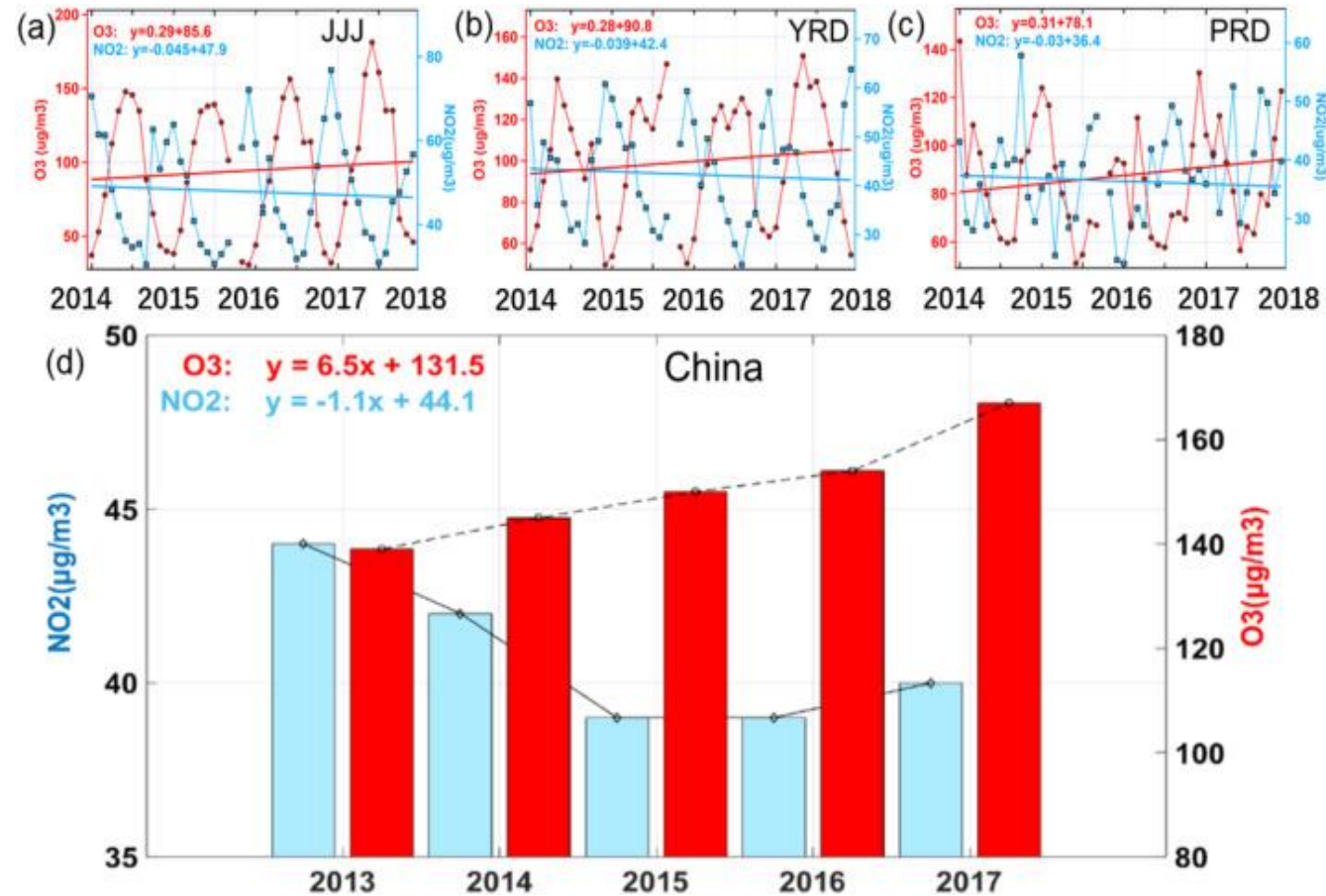
- **Direct forcing:** Ammonium sulfate and ammonium nitrate scatter solar radiation and modify cloud condensation nuclei. N₂O is a potent greenhouse gas.
- Indirect effects: N deposition alters ecosystem productivity and CO₂ uptake; excess deposition accelerates soil N₂O emissions. These net effects remain debated.
- **Climate change is reshaping N_r chemistry:** Warming accelerates NH₃ volatilization and alters photochemical reaction rates.
- **Earth-system frontier:** N_r governance must be embedded in a carbon–nitrogen–climate synergy framework; IPCC AR6 identified N_r cycling as a key uncertainty.

2. Literature Review 5: Nitrogen Cycling in the Pearl River Delta (PRD)



Temporal variations in the percentage of SO_4^{2-} , NO_3^- , NH_4^+ in $\text{PM}_{2.5}$ (Adapted from Yan et al., 2020)

- Total nitrogen deposition in the PRD is rising, with reduced nitrogen (NH_x) now accounting for 60–70%, intensifying coastal eutrophication and biodiversity risk.
- Nitrate dominates autumn–winter haze. Since SO_2 declined sharply (>50%) after 2013, relative NH_3 excess has made nitrate the primary $\text{PM}_{2.5}$ component (25–35% of $\text{PM}_{2.5}$ mass).
- The O_3 paradox: despite ~30% NO_x reduction during 2013–2020, MDA8 O_3 rose ~15 ppb^x, because reduced NO_x weakened the titration sink for O_3 .



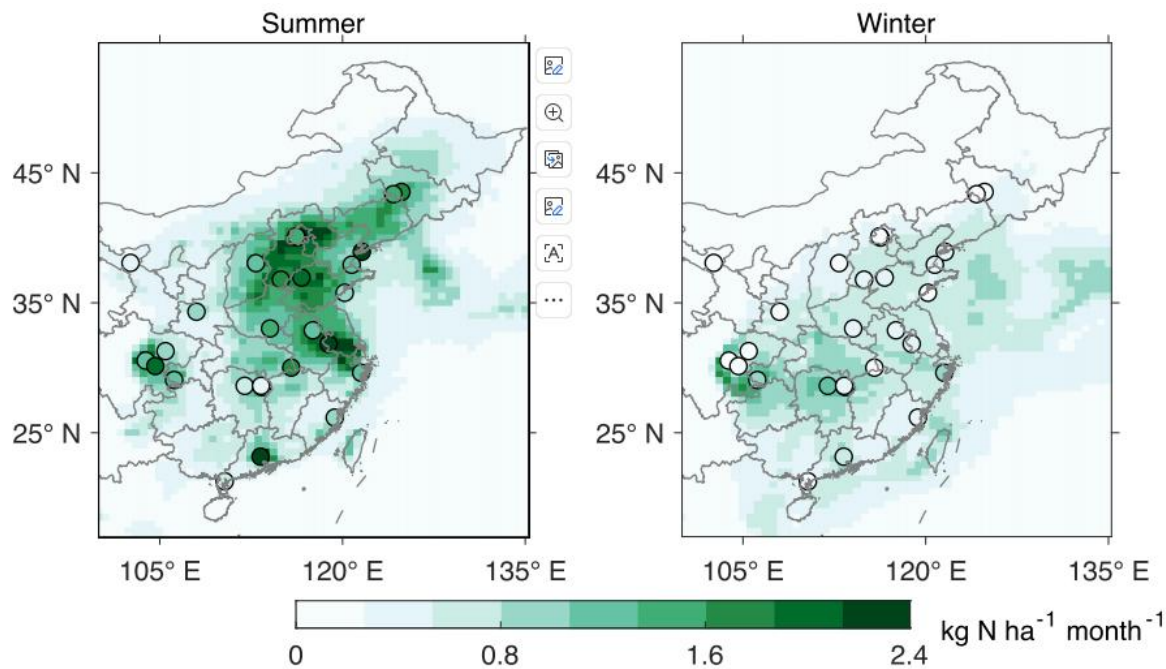
Variations of surface monitored O_3 and NO_2 in Beijing-Tianjin-Hebei (JJJ) (a), Yangtze River Delta (YRD) (b), Pearl River Delta (PRD) (c) and whole China (d) (Adapted from Wang et al., 2019, Figure 4)

2. Literature Review 6: China’s PM_{2.5} Governance Enters a New Stage

污染物	旧标准 (GB 3095—2012)	第一阶段 2026—2030 过渡	第二阶段 2031 年起
PM2.5 年均	35	30	25
PM2.5 日均	75	60	50
SO ₂ 年均	60	60	20
NO ₂ 年均	40	40	30

- March 2026: revised Ambient Air Quality Standards enacted. The Ministry of Ecology and Environment targets Beautiful China 2035 goals and alignment with WHO Interim Target-2 (25 µg m⁻³).
- Primary PM controls are now well-defined; secondary inorganic aerosols require synergistic control of NO_x, VOCs, SO₂, and NH₃, with coordinated governance of PM_{2.5} and O₃.

3. Research Gaps & Motivation: Critical Gaps in PRD Air Quality Management



Spatial distribution of measured (filled circles) and modeled (gridded background) 3-year (2013-2015) averaged summer mean and winter mean nitrate wet deposition fluxes (Zhai et al., 2021).

表1 华北平原与珠江三角洲大气活性氮特征对比

维度	华北平原	珠江三角洲
主导排放源	农业 NH3、燃煤、工业、机动车	交通、工业、港口航运、城市面源、区域输送
气象条件	冬季低温、静稳、边界层低	亚热带高温高湿、海陆风、台风/季风影响明显
活性氮表现形态	颗粒态硝酸盐、铵盐更突出	气态 HNO3/NH3 与 O3 耦合更强，硝酸盐季节性突出
关键化学过程	N2O5 水解、液态水促进硝酸盐增长、气粒分配稳定	光化学氧化、HONO 再生、N2O5/CINO2、海盐/航运影响
主要污染表征	秋冬季 PM2.5 与硝酸盐主导	秋季 O3 突出，冬季/过渡季硝酸盐增强
氮沉降特征	干沉降相对重要，农田氮负荷高	湿沉降更强，河口/近海富营养化风险高

- Geographic bias: Existing NH₃ research concentrates on the North China Plain. The PRD differs fundamentally: subtropical high temperatures and humidity drive extremely active photochemistry and rapid secondary aerosol formation. Yet systematic long-term NH₃ observations in the PRD are virtually absent.
- Composition shift: Although PRD PM_{2.5} mass meets standards, its composition has shifted dramatically. Sulfate has fallen, but nitrate now drives autumn–winter haze. The specific mechanism by which excess NH₃ fuels this nitrate enhancement remains unresolved.
- O₃–PM_{2.5} ternary coupling: Despite deep NO_x cuts, O₃ continues to rise. The ternary coupling mechanism of NH₃–NO_x–VOCs under the latest emission trends is still poorly understood.

4. Research Objectives

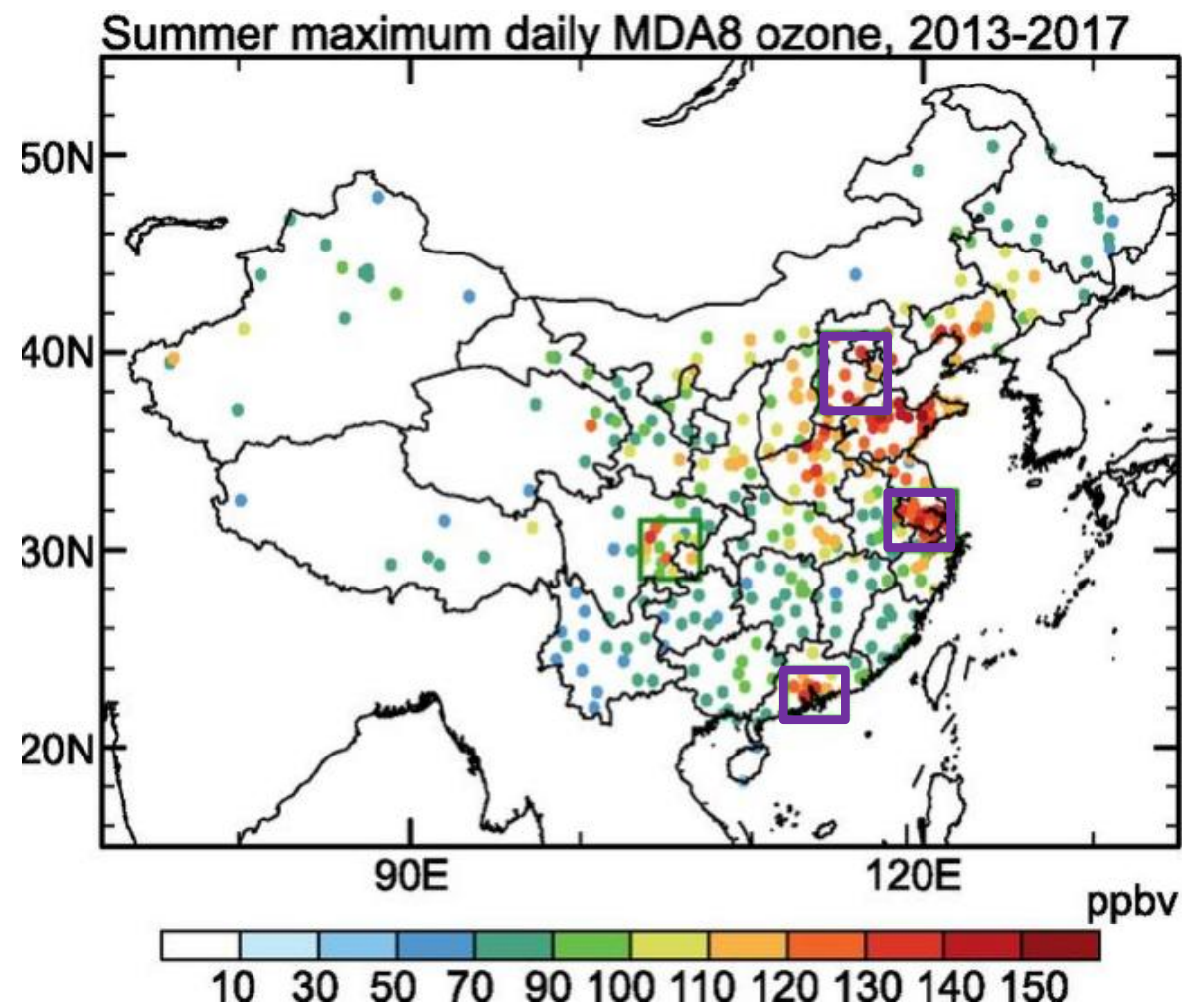
To understand the chemical evolution of atmospheric reactive nitrogen in the PRD and its impacts on $\text{PM}_{2.5}$ composition, this dissertation is structured around four pillars:

1. To Detect the long-term evolution, spatial differentiation, and structural transitions of anthropogenic SO_2 , NO_x , and NH_3 emissions in the PRD.
2. To demonstrate how changes in reactive nitrogen species affect $\text{PM}_{2.5}$ chemical composition.
3. To evaluate atmospheric nitrogen deposition fluxes and their ecological/environmental effects in the PRD.
4. To project the future emission-reduction scenarios and synergistic $\text{PM}_{2.5}$ – O_3 control pathways.

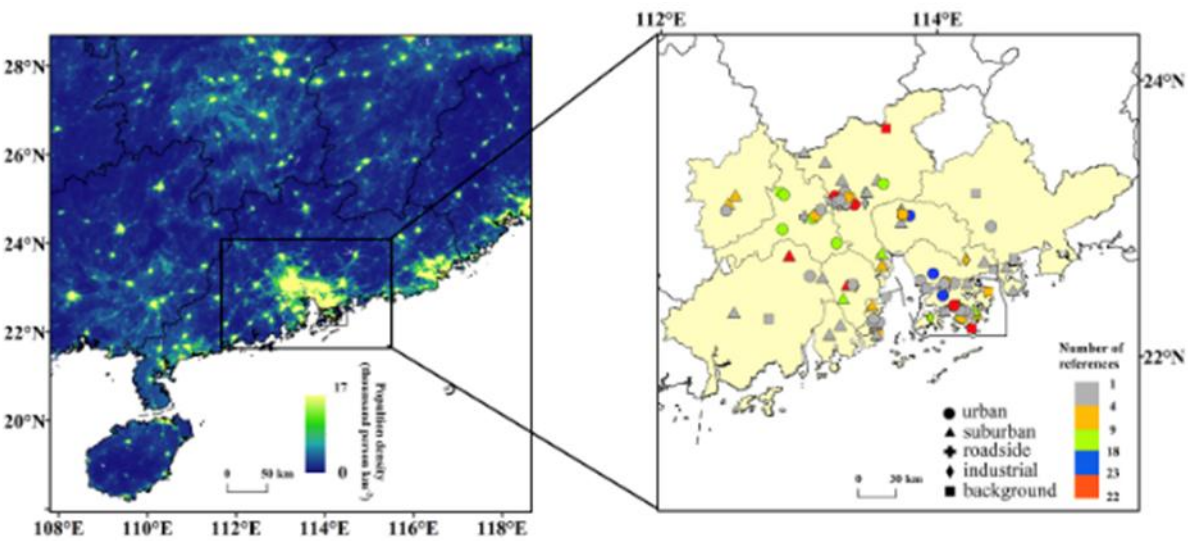
Dissertation Structure

- Chapter 1: Introduction
- Chapter 2: Data Sources and Methodology
- Chapter 3: Long-term Evolution and Structural Transition of SO₂, NO_x, and NH₃ Emissions in the PRD
- Chapter 4: Impacts of Reactive Nitrogen Changes on PM_{2.5} Composition
- Chapter 5: Atmospheric Nitrogen Deposition Fluxes and Eco-environmental Effects
- Chapter 6: Future Reduction Scenarios and Synergistic Governance Pathways
- Chapter 7: Conclusions and Outlook

5. Data and Methodology–Study Area



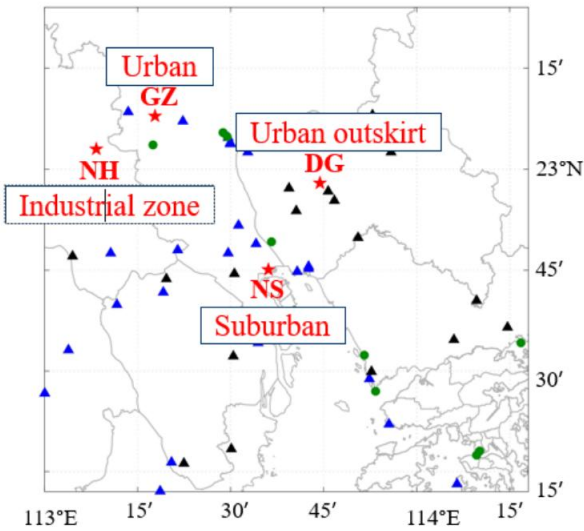
Study area from Li et al., 2019.



Feature	Pearl River Delta (PRD)	Yangtze River Delta (YRD)	Beijing–Tianjin–Hebei (BTH)
geographical scope	21.5°–24°N, 112°–115.5°E	30°–33°N, 118°–122°E	37°–41°N, 114°–118°E
Climate Type	Subtropical monsoon climate	Subtropical monsoon climate	Temperate monsoon climate
Annual Temperatur (°C)	22–23	16–18	12–14
Annual Precipitatio (mm)	1600–2000	1100–1500	500–700

5. Data and Methodology–Data

- **Emission Inventory:** Multi-resolution Emission Inventory for China (MEIC).
- Period: 1990–2023.
- Pollutants: NO_x, SO₂ , and NH₃ .
- Spatial scale: Gridded anthropogenic emissions.
- Study regions:
 - Pearl River Delta (PRD);
 - Beijing–Tianjin–Hebei (BTH);
 - Yangtze River Delta (YRD).



Site	Represent	Period
GZ (Guangzhou)	Urban	20070724-20171227
NS (Nanshan)	Suburban	20070724-20171227
NH (Nanhai)	Industrial zone	20080114-20141227; 2017
DG (Dongguan)	Urban outskirts	20080706-20141127

Locations of observed sites

Reanalyse dataset: MERRA2
Emissions: CEDS; MEIC
Resolution: 0.5x0.625 nested grid
Period: 2015-2020 monthly

- **PM2.5 data:** Observed PM2.5 and its species from Prof. Jianzheng Yu
- Site: GZ, NS, NH and DG station.
- Timestamp: 2007–2017.
- Pollutants: PM2.5, NH₄⁺, NO₃⁻, SO₄²⁻, Cl⁻, OC, EC and others

- **NNDMN Dataset:** Nationwide Nitrogen Deposition Monitoring Network.
- Site: Baiyun (23.16 ° N, 113.27 ° E)
- Period: May 2010–Dec. 2020.
- Pollutants: NH₃, NO₂, HNO₃, NH₄⁺, NO₃⁻.

5. Data and Methodology–Methodology

Introduction of GOES-Chem

The GEOS – Chem model is a global 3-D model of atmospheric composition driven by assimilated meteorological observations from the Goddard Earth Observing System (GEOS) of the NASA Global Modeling and Assimilation Office. It is applied by research groups around the world to a wide range of atmospheric composition problems. Central management and support of the model is provided by the Atmospheric Chemistry Modeling Group at Harvard University and by the Atmospheric Composition Analysis Group at Washington University.

模式配置

- 气象模式：三层嵌套 27→9→3 km，最内层覆盖 PDR；
- Resolution: 0.5x0.625 nested grid
- 化学模式： CMAQ v5.3.3；
- 气相化学机理： SAPRC07TC；
- 气溶胶模块： AERO7 + ISORROPIA-II；
- 多相化学： 打开卤素化学（Halogen chemistry）与 N_2 O_5 / ClNO_2 非均相参数化；
- 排放输入： MEIC + SMOKE 时空分配；生物源使用 MEGAN v3.1；船舶源使用 CAMS-GLOB-SHIP；
- 模拟时段： Jan 2017—Dec. 2017

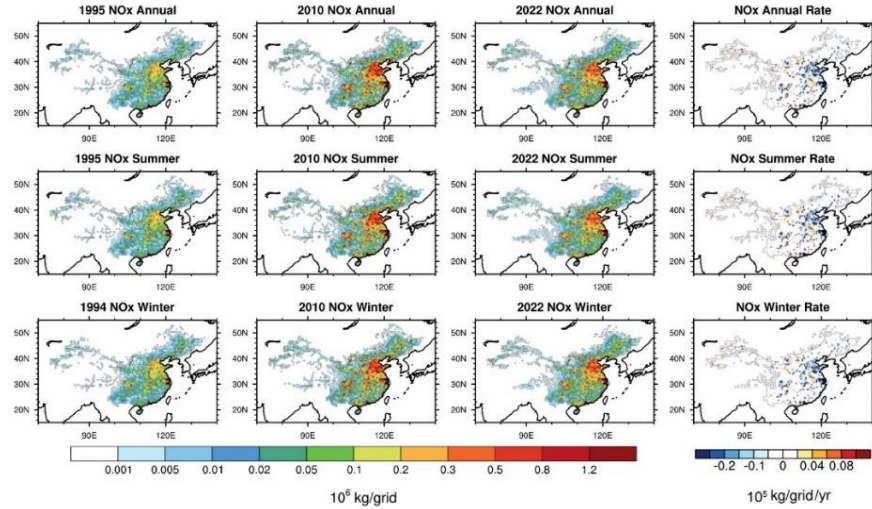
6. Preliminary Results — Chapter 3 Overview

Title: Long-term Evolution and Structural Transition of SO₂, NO_x, and NH₃ Emissions in the PRD

- Objective: Reveal long-term emission patterns, source-structure transitions, and spatial coupling of reactive nitrogen and acidic precursors in the PRD.
- Research Components:
 1. Spatiotemporal changes of NO_x, SO₂, and NH₃ in China (1990–2023)
 2. Spatiotemporal changes of NO_x, SO₂, and NH₃ in the PRD (1990–2023)
 3. Source-structure evolution in the PRD
 4. Changes in oxidized vs. reduced nitrogen emission structure
 5. Comparative emission transition pathways among PRD, BTH, and YRD

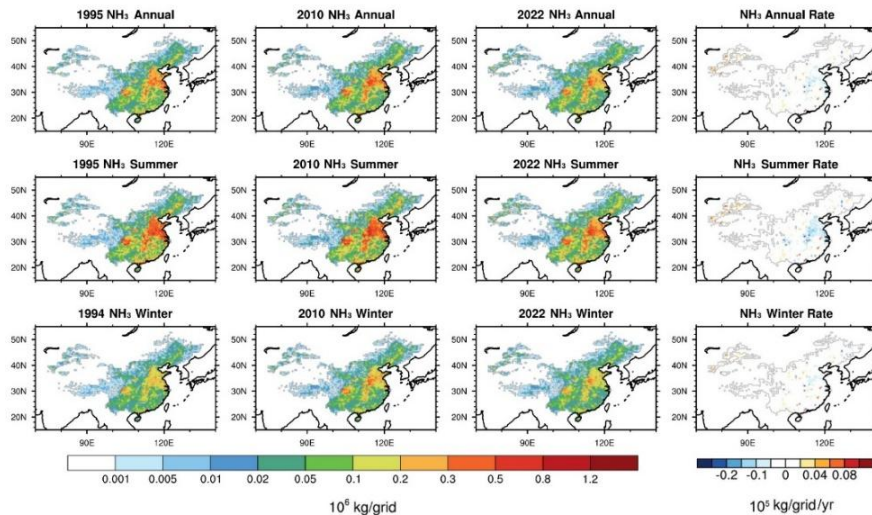
6. Preliminary Results 1.1: National Spatiotemporal Trends of NO_x, SO₂, and NH₃ (1990–2023)

NO_x Emission and Annual Change Rate

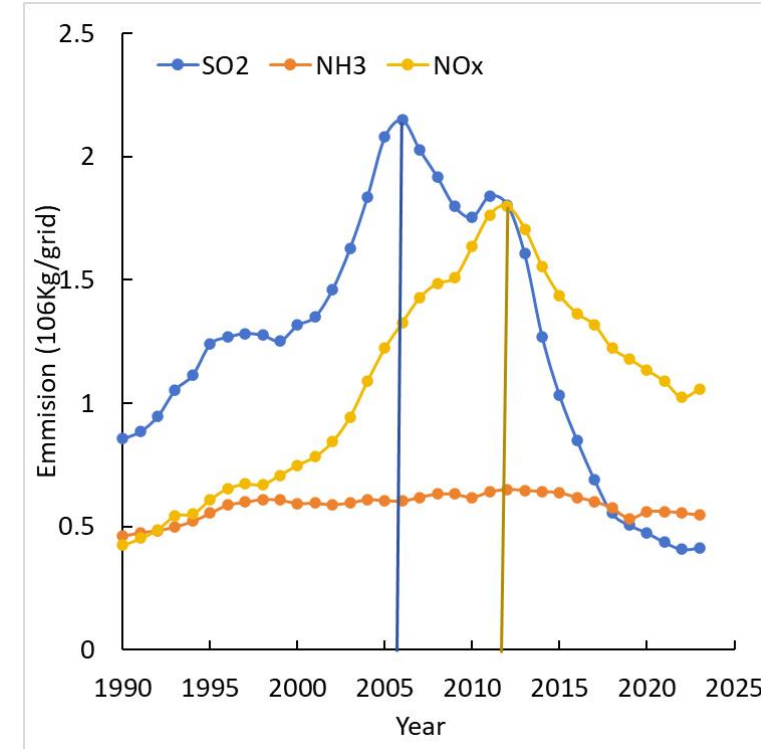


Spatial distribution of annual, summer and winter NO_x emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in China (10⁶kg/month/grid)

NH₃ Emission and Annual Change Rate



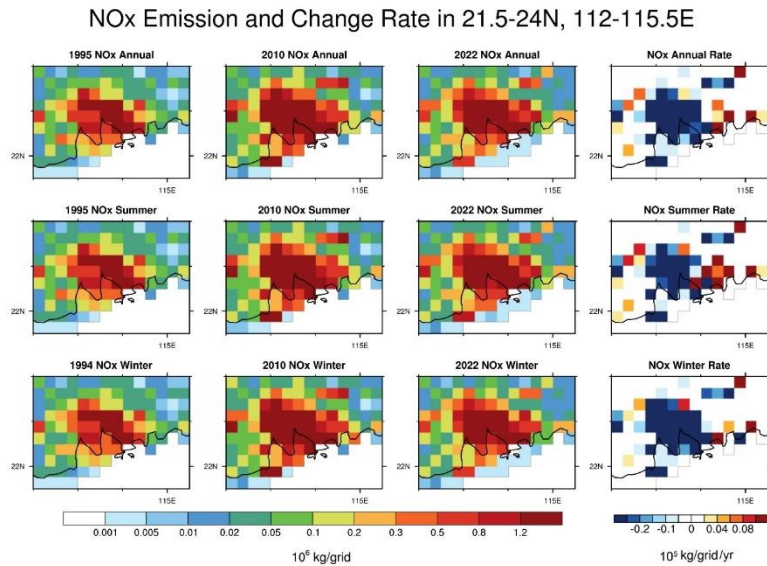
Spatial distribution of annual, summer and winter NH₃ emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in China (10⁶kg/month/grid)



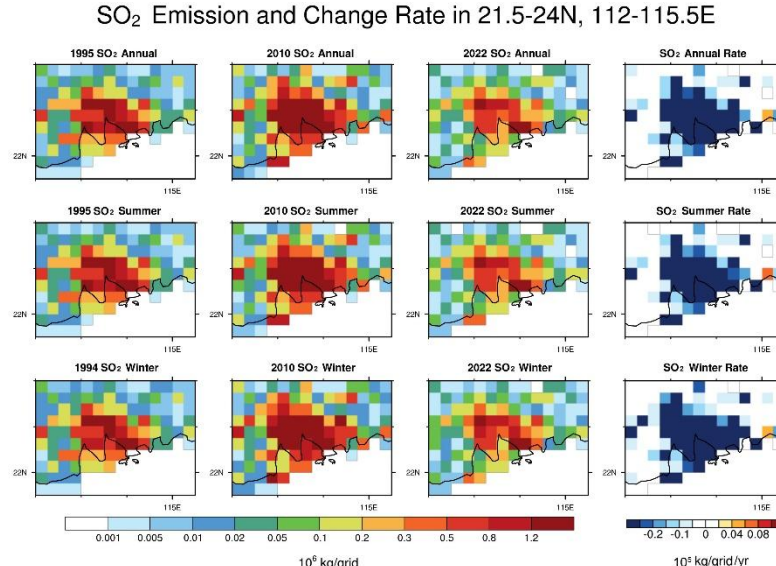
The annual emission intensity of NO_x, SO₂, and NH₃ in China in 1995, 2010, and 2022.

- NO_x and SO₂ emissions exhibited inverted-U trends, peaking in 2006 and 2012, respectively, reflecting energy growth, industrial expansion, and subsequent strengthened emission controls.
- NH₃ emissions, driven by agricultural structure and temperature-dependent volatilization, remained comparatively stable, increasing the relative abundance of NH₃ in the national Nr budget.

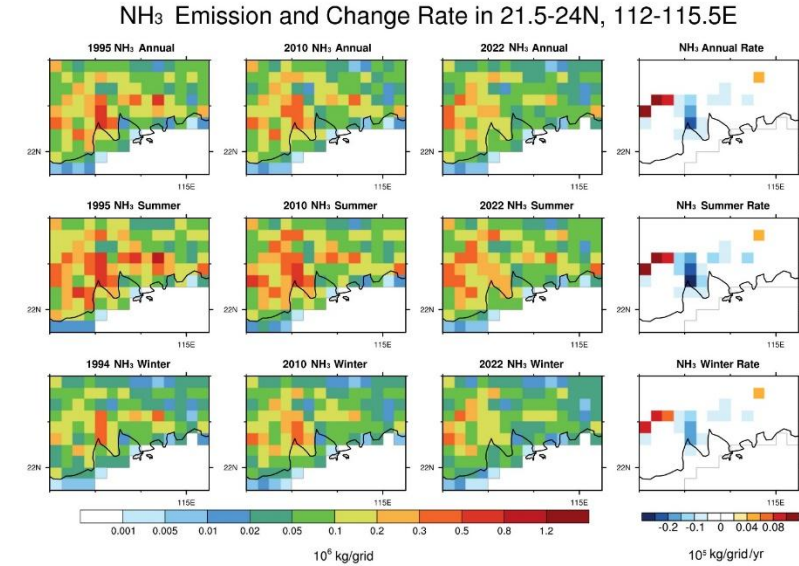
6. Preliminary Results 1.2: PRD Spatiotemporal Trends of NO_x, SO₂, and NH₃ (1990–2023)



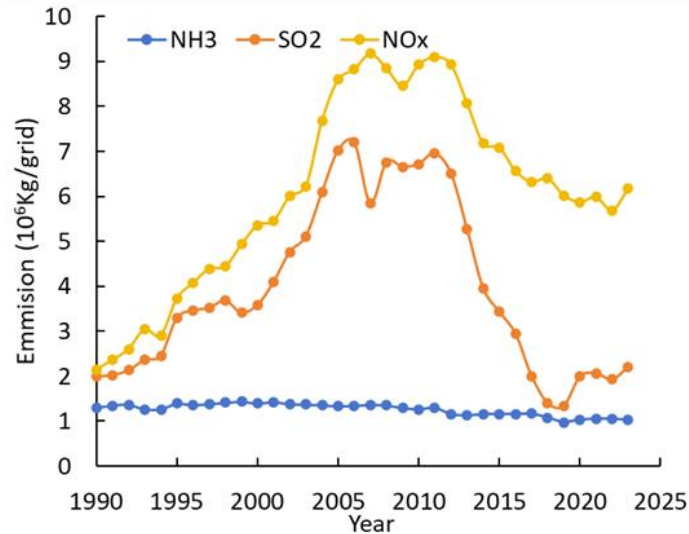
Spatial distribution of annual, summer and winter NO_x emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in PRD (10⁶kg/month/grid))



Spatial distribution of annual, summer and winter SO₂ emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in PRD (10⁶kg/month/grid))



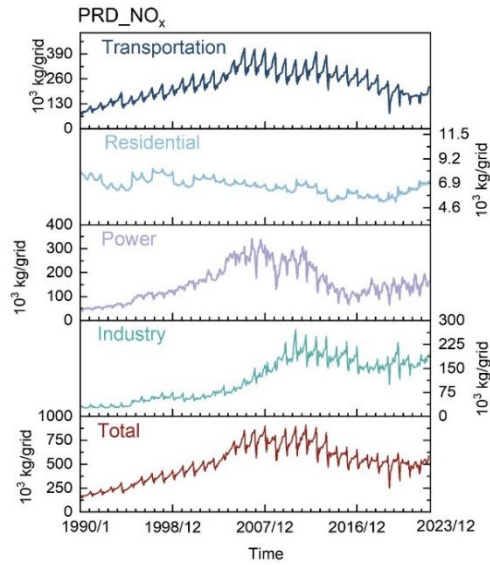
Spatial distribution of annual, summer and winter NH₃ emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in PRD (10⁶kg/month/grid))



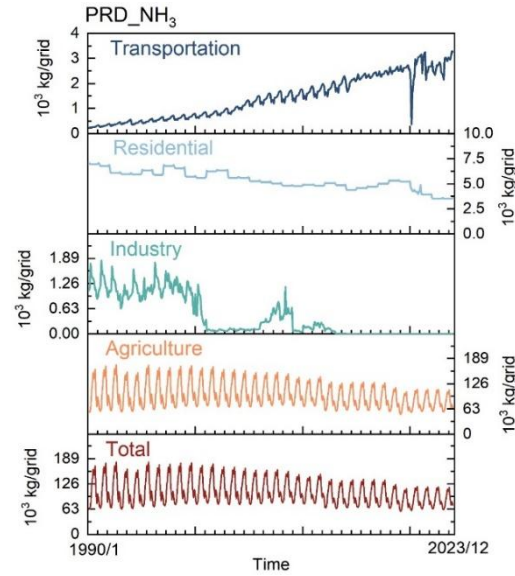
Changes in average NO_x, SO₂, and NH₃ emission intensity in the PRD from 1990 to 2023.

- PRD NO_x and SO₂ emissions peaked in 2006 and 2005, respectively—earlier than the national average—reflecting earlier industrialization and aggressive local governance.
- The reduced-to-oxidized nitrogen ratio increased; the regional Nr structure shifted from NO_x-dominance toward enhanced NH₃ importance.
- Spatial evolution: from 1990s point-source dominance → 2010 urban-area diffusion → 2022 significantly reduced overall intensity. However, the Guangzhou–Foshan and Shenzhen–Dongguan metropolitan belts remain emission hotspots.
- Spatial misalignment: Post-2010, NH₃ intensity weakened in city centers but increased in peripheral agricultural areas (Zhaoqing, Jiangmen), creating a pronounced urban–rural gradient. This decoupling provides a coupled-precursor foundation for downwind ammonium nitrate formation.

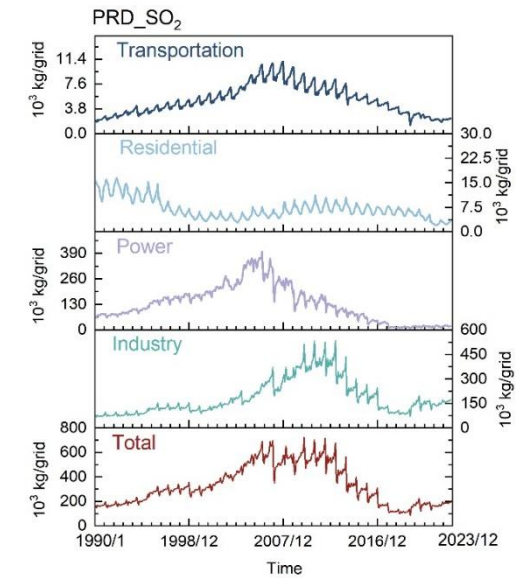
6. Preliminary Results 1.3: Source-Structure Evolution of NO_x, SO₂, and NH₃ in the PRD (1990–2023)



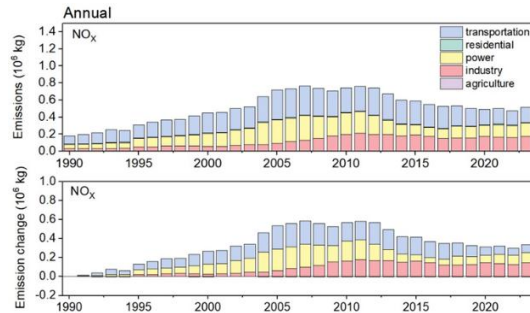
Monthly changes in NH₃ emission intensity from different sources in the PRD region from 1990 to 2023



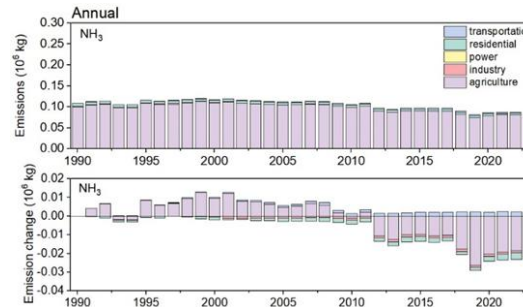
Monthly changes in NH₃ emission intensity from different sources in the PRD region from 1990 to 2023



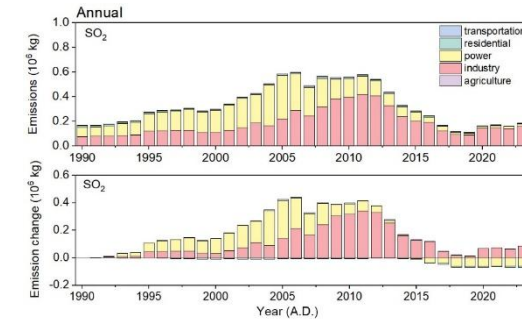
Monthly changes in SO₂ emission intensity from different sources in the PRD region from 1990 to 2023



Changes in NO_x emission intensity from different sources in the PRD from 1990 to 2023, and changes in NO_x emission intensity from different sources compared to 1990.



Changes in NH₃ emission intensity from different sources in the PRD from 1990 to 2023, and changes in NO_x emission intensity from different sources compared to 1990.



Changes in SO₂ emission intensity from different sources in the PRD from 1990 to 2023, and changes in NO_x emission intensity from different sources compared to 1990.

- Changes in emission structure determine the chemical constraint state of the sulfate–nitrate–ammonium system.
- NO_x: Dominated by traffic, power, and industry. Pre-2012 increases were driven by traffic and power; post-2012 declines were also led by traffic (emission-standard upgrades).
- NH₃: Agricultural sector dominates (>92% of total). High summer temperatures exacerbate volatilization, causing pronounced summer emission peaks.
- SO₂: Early dominance by power and industry. Power-sector SO₂ dropped ~94% from its 2006 peak due to fuel switching and FGD penetration; industrial SO₂ fell 61% from its 2011 peak but now constitutes >87% of the remaining total.

6. Preliminary Results 1.4: Changes in Oxidized vs. Reduced Nitrogen Emission Structure in the PRD

NO_x , SO_2 , and NH_3 collectively determine precursor availability for the sulfate–nitrate–ammonium system.

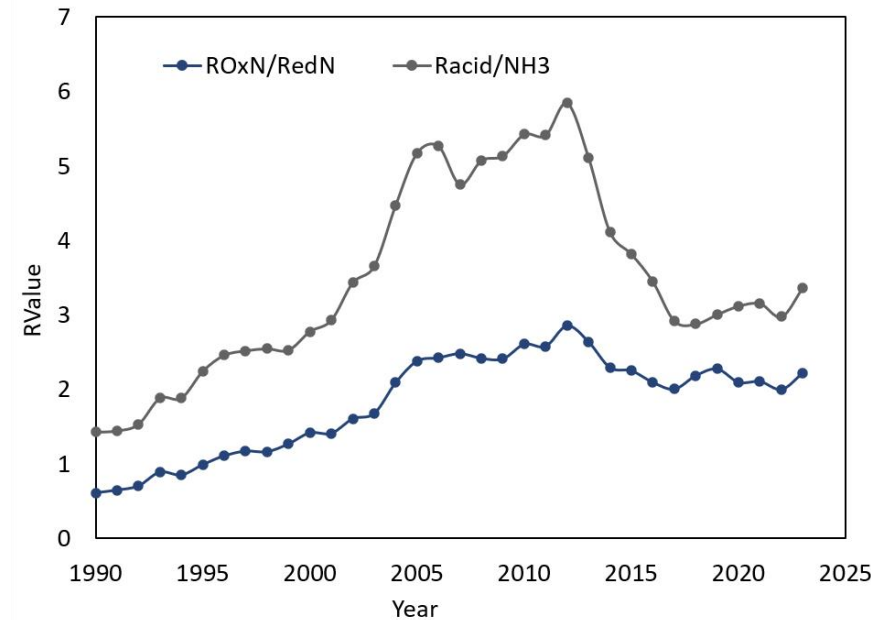
To compare emissions on a nitrogen-element basis.

Oxidized/Reduced Nitrogen Ratio:

$$R_{\text{OxN/RedN}} = \frac{E_{\text{OxN}}}{E_{\text{RedN}}} = \frac{E_{\text{NO}_x}}{E_{\text{NH}_3}} \times \frac{17}{46}$$

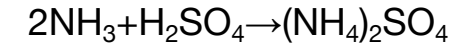
Potential Acid Precursor / Ammonia Molar Ratio:

$$R_{\text{acid/NH}_3} = \frac{2 \frac{E_{\text{SO}_2}}{64} + \frac{E_{\text{NO}_x}}{46}}{E_{\text{NH}_3}/17}$$



Changes in the ratio of oxidized/reduced nitrogen emissions ($R_{\text{OxN/RedN}}$) and oxidized/reduced nitrogen emissions ($R_{\text{acid/NH}_3}$) from 1990 to 2023 in PRD

NH_3 preferentially neutralizes the strongest acid, H_2SO_4 (irreversible):

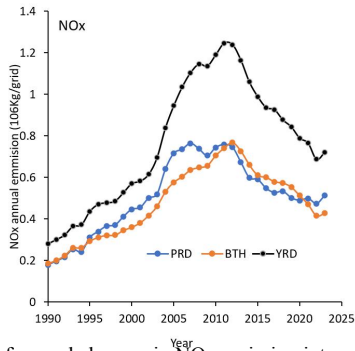


any remaining NH_3 establishes a semi-volatile equilibrium with HNO_3 :

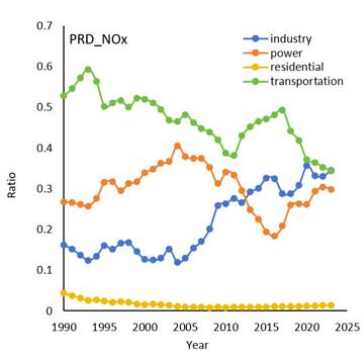


- **1990–2012:** $R_{\text{OxN/RedN}}$ rose from 0.6 to 2.86 (NO_x grew faster than NH_3).
- **2012–2023:** $R_{\text{OxN/RedN}}$ declined slowly (NO_x fell faster than NH_3), increasing the relative share of reduced nitrogen.
- $R_{\text{acid/NH}_3}$ declined post-2012, indicating enhanced NH_3 supply relative to acidic precursors.
- **Hypothesis:** As SO_2 and NO_x decline faster than NH_3 , the PRD system is becoming increasingly NH_3 -rich. This may elevate nitrate dependence on NO_x oxidation products, meteorology, and regional transport. This hypothesis will be tested in subsequent chapters via observations and chemical-transport sensitivity experiments.

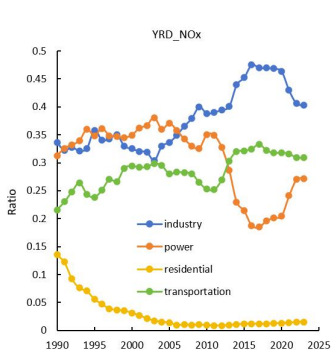
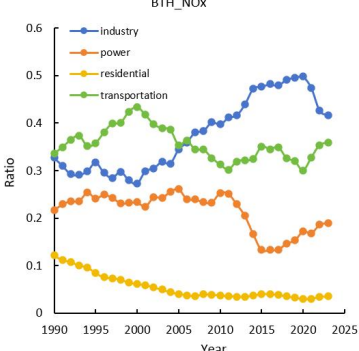
6. Preliminary Results 1.5: Comparative Emission Transition Pathways: PRD, BTH, and YRD



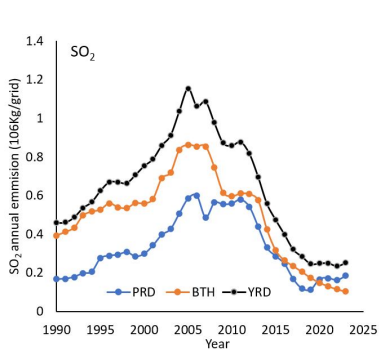
Comparison of annual changes in NOx emission intensity among the three regions from 1990 to 2023



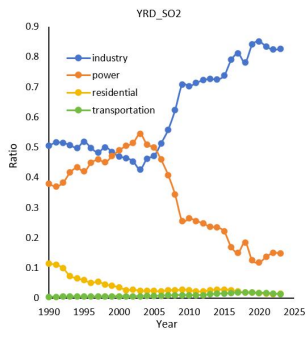
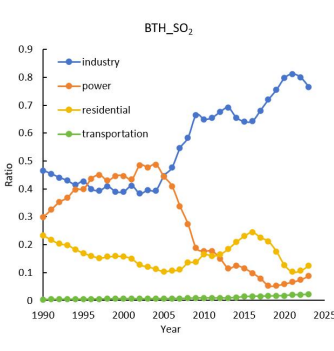
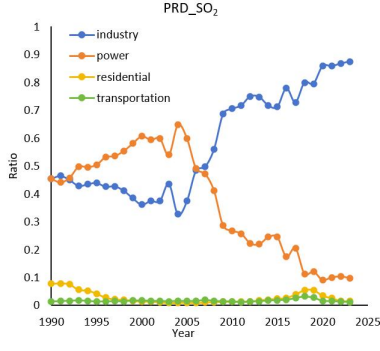
Comparison of NOx emission intensity changes from different sources in the three regions from 1990 to 2023



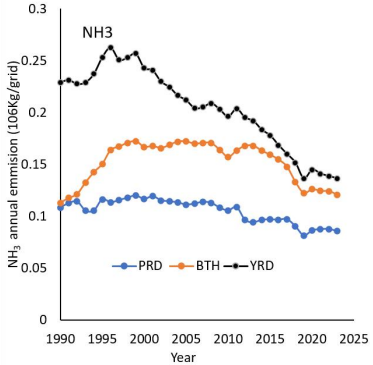
- All three clusters show rapid growth–peak–decline patterns for NO_x and SO₂; SO₂ declines were most pronounced in BTH.
- NH₃ remains agriculturally dominated across all three, with slow declines.
- YRD: Highest overall emission intensity; mixed industrial, power, traffic, and agricultural sources.
- BTH: Historically high industrial shares of NO_x and SO₂; SO₂ governance most successful.
- PRD: Historically traffic-dominant NO_x; evolving toward a balanced industrial–traffic–power structure.
- Implication: SO₂ governance focus has shifted from power to industry; NO_x governance must address industry, traffic, and power simultaneously. Region-specific, multi-pollutant, multi-sector strategies are required.



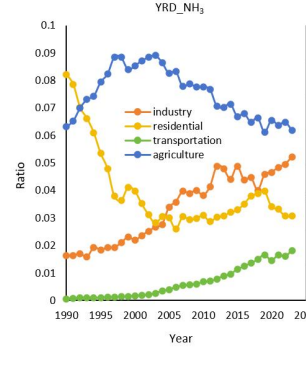
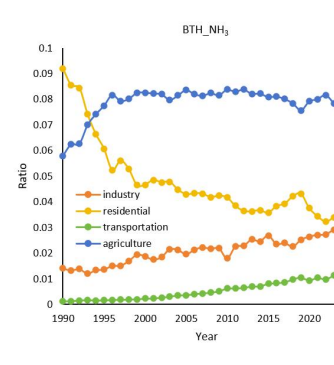
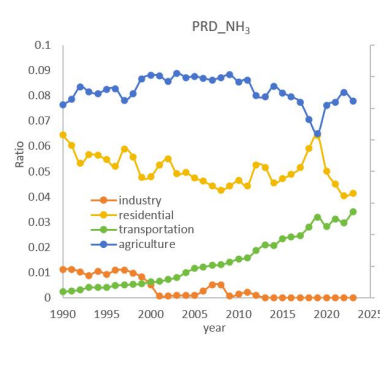
Comparison of annual changes in SO2 emission intensity among the three regions from 1990 to 2023



Comparison of SO2 emission intensity changes from different sources in the three regions from 1990 to 2023



Comparison of annual changes in NH3 emission intensity among the three regions from 1990 to 2023



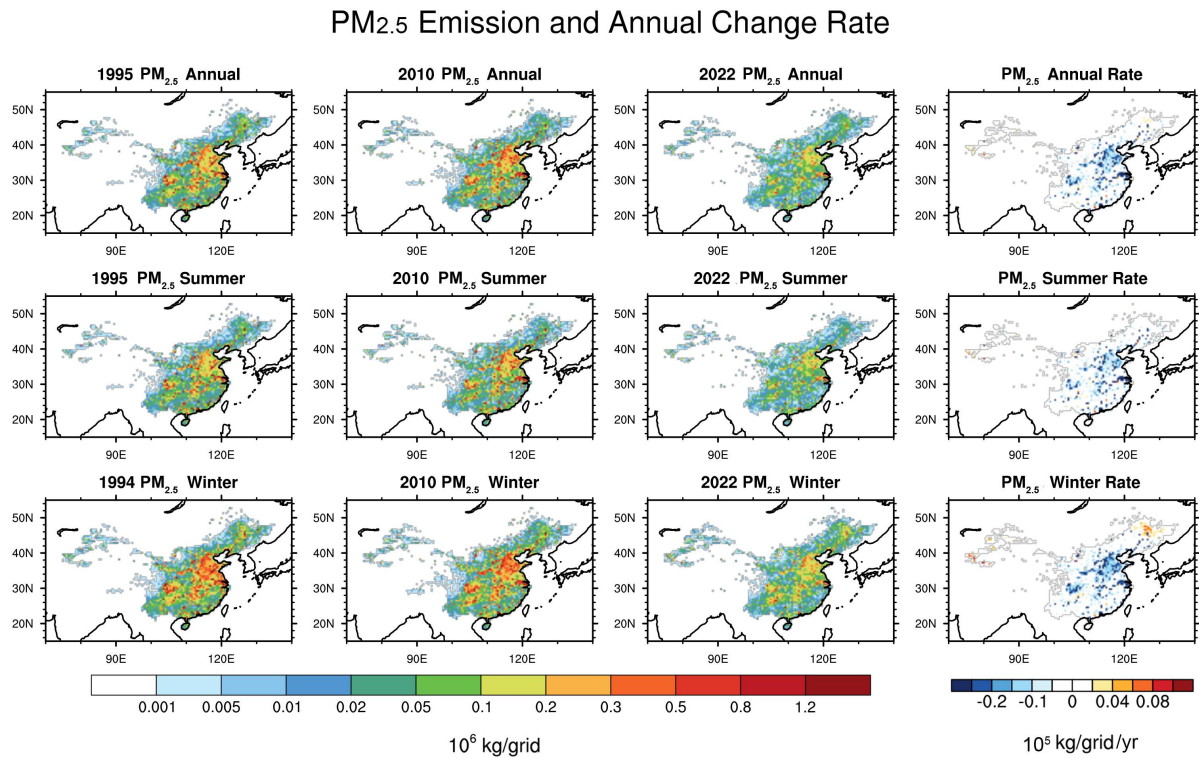
Comparison of NH3 emission intensity changes from different sources in the three regions from 1990 to 2023

6. Preliminary Results – Chapter 4 Overview

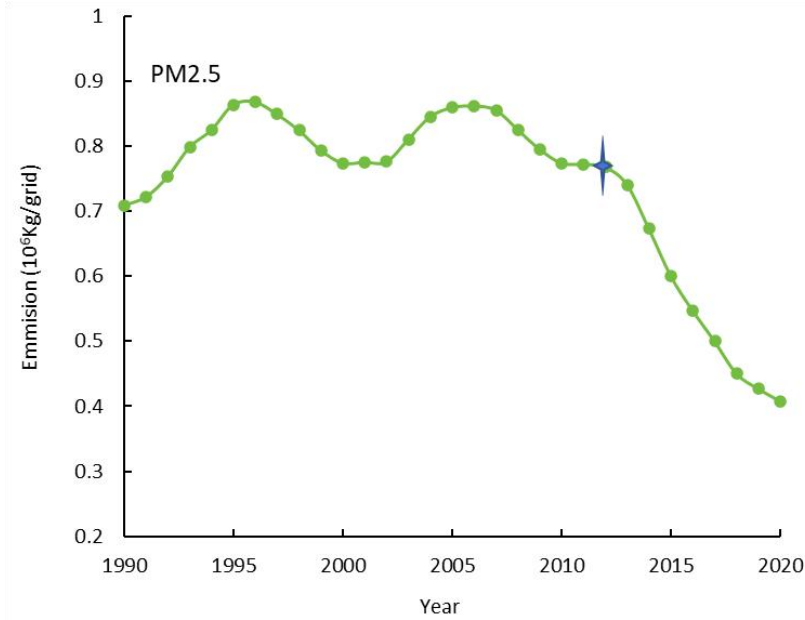
Title: Impacts of Reactive Nitrogen Changes on PM_{2.5} Composition

- Objective: Reveal long-term evolution of anthropogenic PM_{2.5} emissions and source-structure transition in the PRD. Identify how changes in oxidized nitrogen, reduced nitrogen, and sulfur precursors affect PM_{2.5} composition, and understand the nonlinear PM_{2.5} response to NO_x and NH₃ changes.
- Research Components:
 1. Spatiotemporal characteristics of PM_{2.5} emissions in China (1990 – 2023)
 2. Spatiotemporal characteristics and source–structure evolution of PM_{2.5} in the PRD
 3. PM_{2.5} component changes in the PRD (1990 – 2023)
 4. GEOS–Chem simulated PM_{2.5} component changes
 5. Nonlinear mechanism of PM_{2.5} response to NO_x and NH₃ changes

6. Preliminary Results 2.1: National Spatiotemporal Trends of PM_{2.5} Emissions (1990–2023)



Spatial distribution of annual, summer and winter PM_{2.5} emission intensity in 1995,2010 and 2022, and the rate of change from 2010 to 2022 in China (106kg/month/grid))

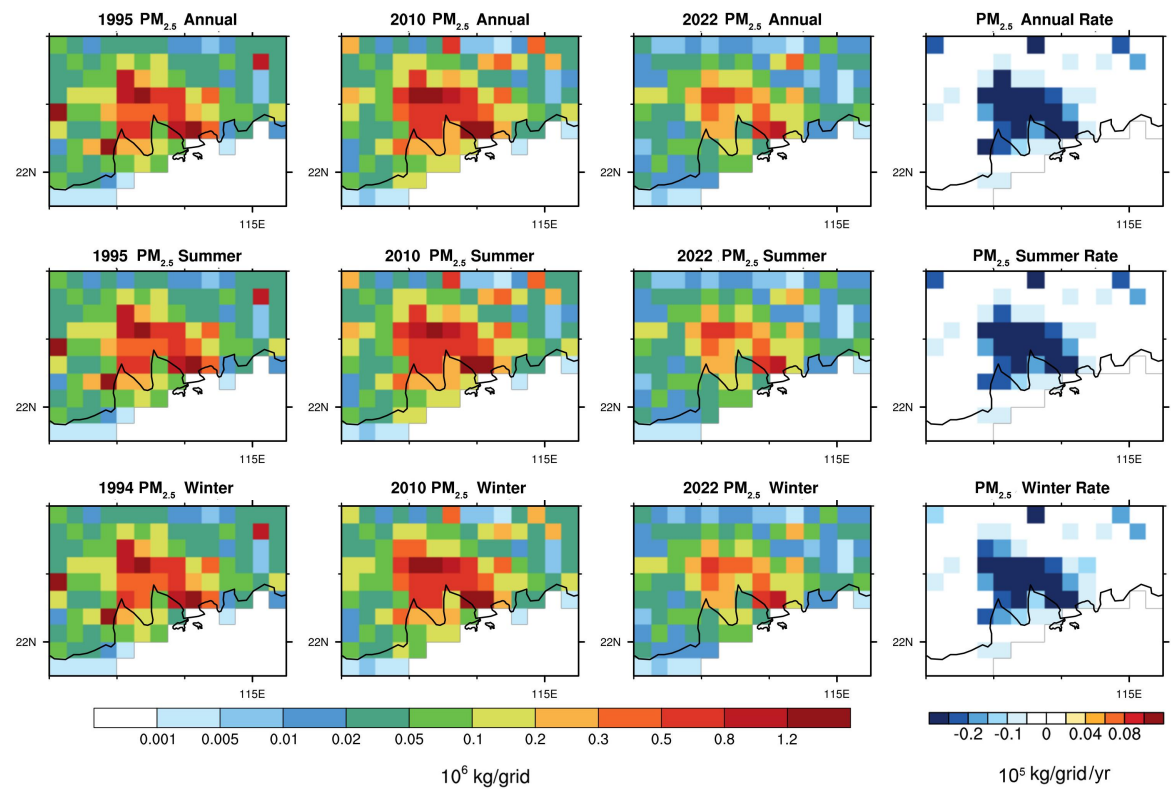


The annual emission intensity of PM_{2.5} in China in 1995, 2010, and 2022, respectively.

- National PM_{2.5} emission intensity peaked in 2006, declining sharply after 2012 due to strengthened controls.
- From 2010 to 2023, PM_{2.5} reductions were concentrated in North China, the Sichuan Basin, the middle–lower Yangtze River, and the PRD.

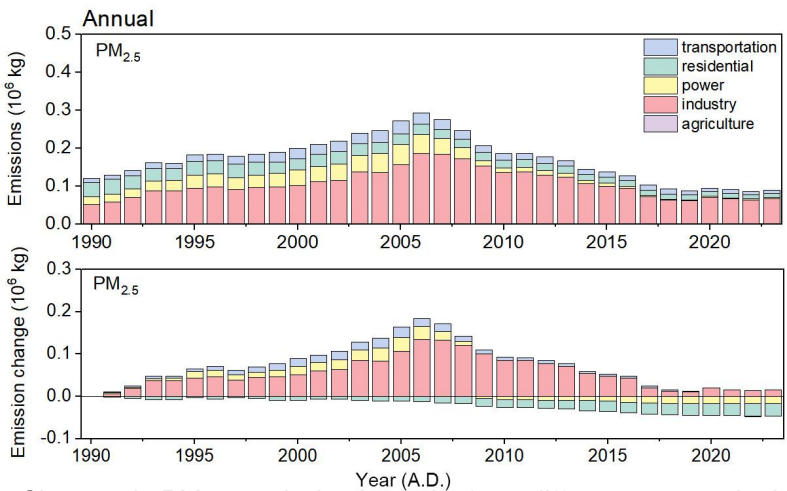
6. Preliminary Results 2.2: PRD Spatiotemporal Trends and Source-Structure Evolution of PM_{2.5} (1990–2023)

PM_{2.5} Emission and Change Rate in 21.5-24N, 112-115.5E

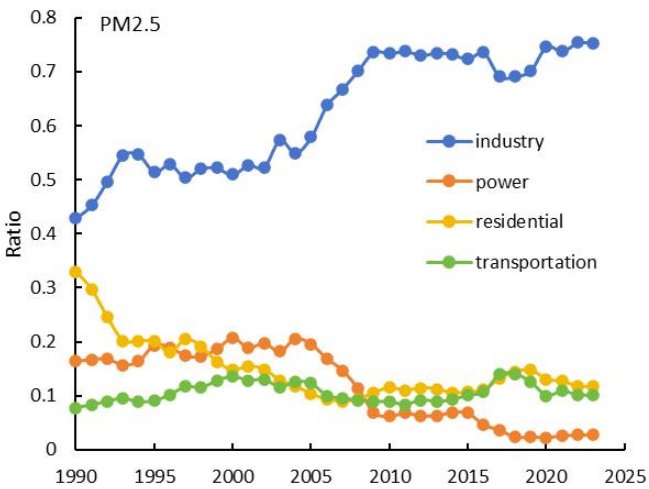


Spatial distribution of annual, summer and winter PM_{2.5} emission intensity in 1995, 2010 and 2022, and the rate of change from 2010 to 2022 in PRD (10⁶kg/month/grid)

- PRD PM_{2.5} emission intensity peaked in 2006, consistent with the national trend.
- From 2010 to 2022, overall intensity dropped significantly, yet Guangzhou–Foshan and Shenzhen–Dongguan remain high-emission zones.
- Industrial PM_{2.5} share rose from 42% (1990) to 75% (2023), while power and residential shares declined continuously.

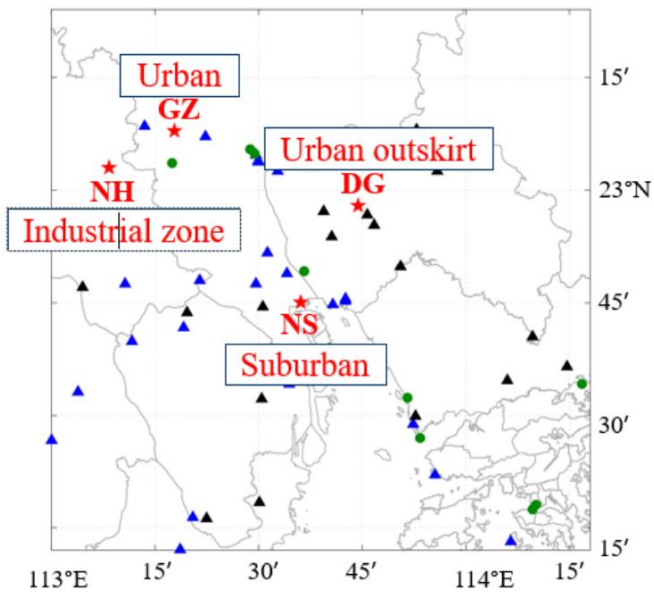


Changes in PM_{2.5} emission intensity from different sources in the PRD from 1990 to 2023, and changes in PM_{2.5} emission intensity from different sources compared to 1990.

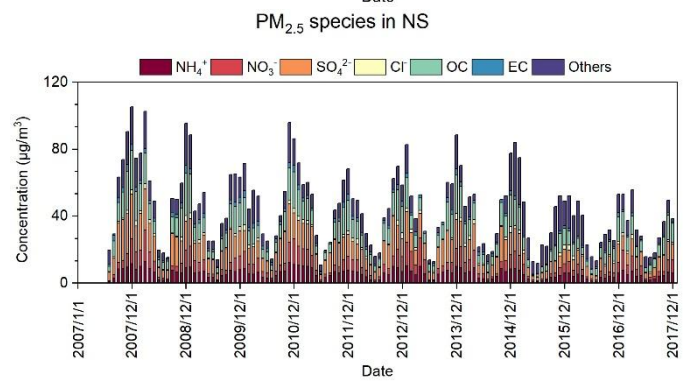
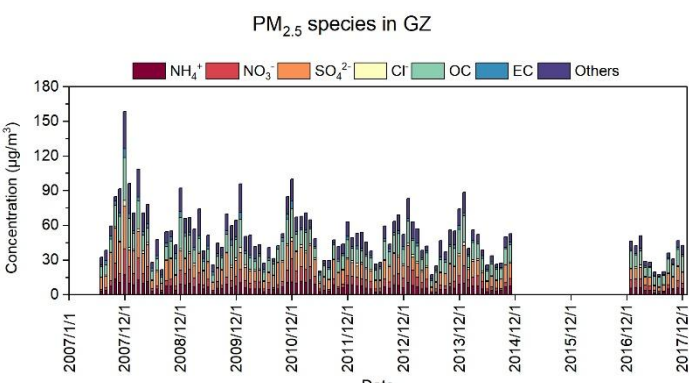
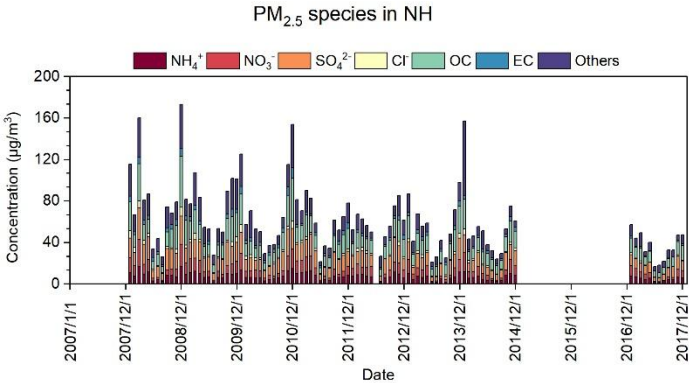
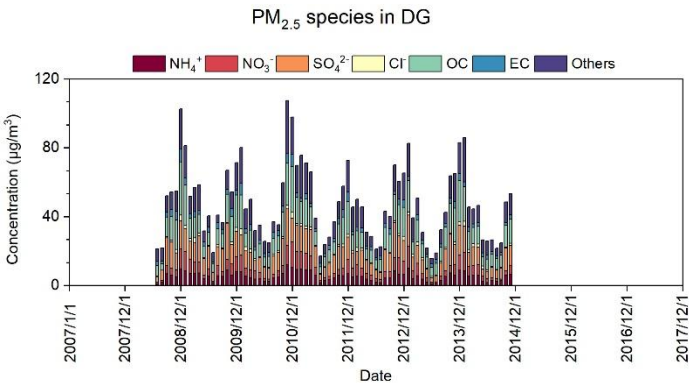


Percentage of PM_{2.5} emissions from different sources in PRD from 1990 to 2023

6. Preliminary Results 2.3: Observational Evidence of PM_{2.5} Component Changes in the PRD (2007–2017)

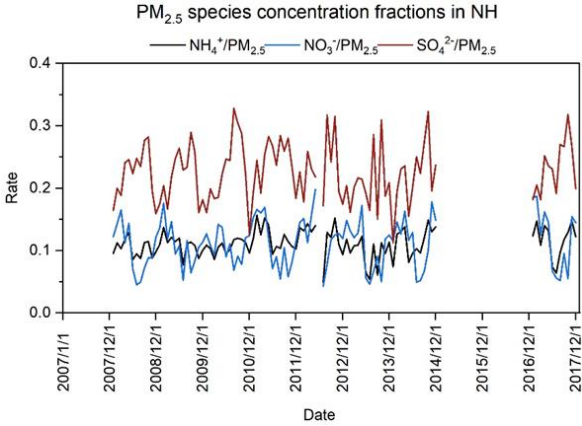


Observation sites in PRD region

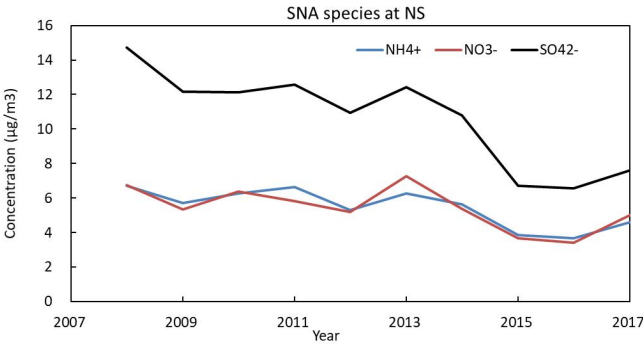


- Analysis of four monitoring sites shows PM_{2.5} mass decreased during 2007–2017. Industrial sites (NH) recorded the highest concentrations; suburban sites (DG) the lowest.
- NH₃ and NH₄⁺ exhibited distinct winter-high, summer-low seasonality.
- SNA (SO₄²⁻ + NO₃⁻ + NH₄⁺) averaged 43% of PM_{2.5} at DG (up to 50% in summer) and 50% at NH (up to 60% in summer), indicating that summer PM_{2.5} in the PRD is largely driven by secondary inorganic aerosol formation.
- Seasonal asymmetry: NO₃⁻ contribution was low in summer (~5%) but high in winter (~15%); SO₄²⁻ showed the opposite pattern (summer ~30%, winter ~15%).

2007–2017年DG, GZ, NH和NS四个观测点PM_{2.5}及其组分浓度变化

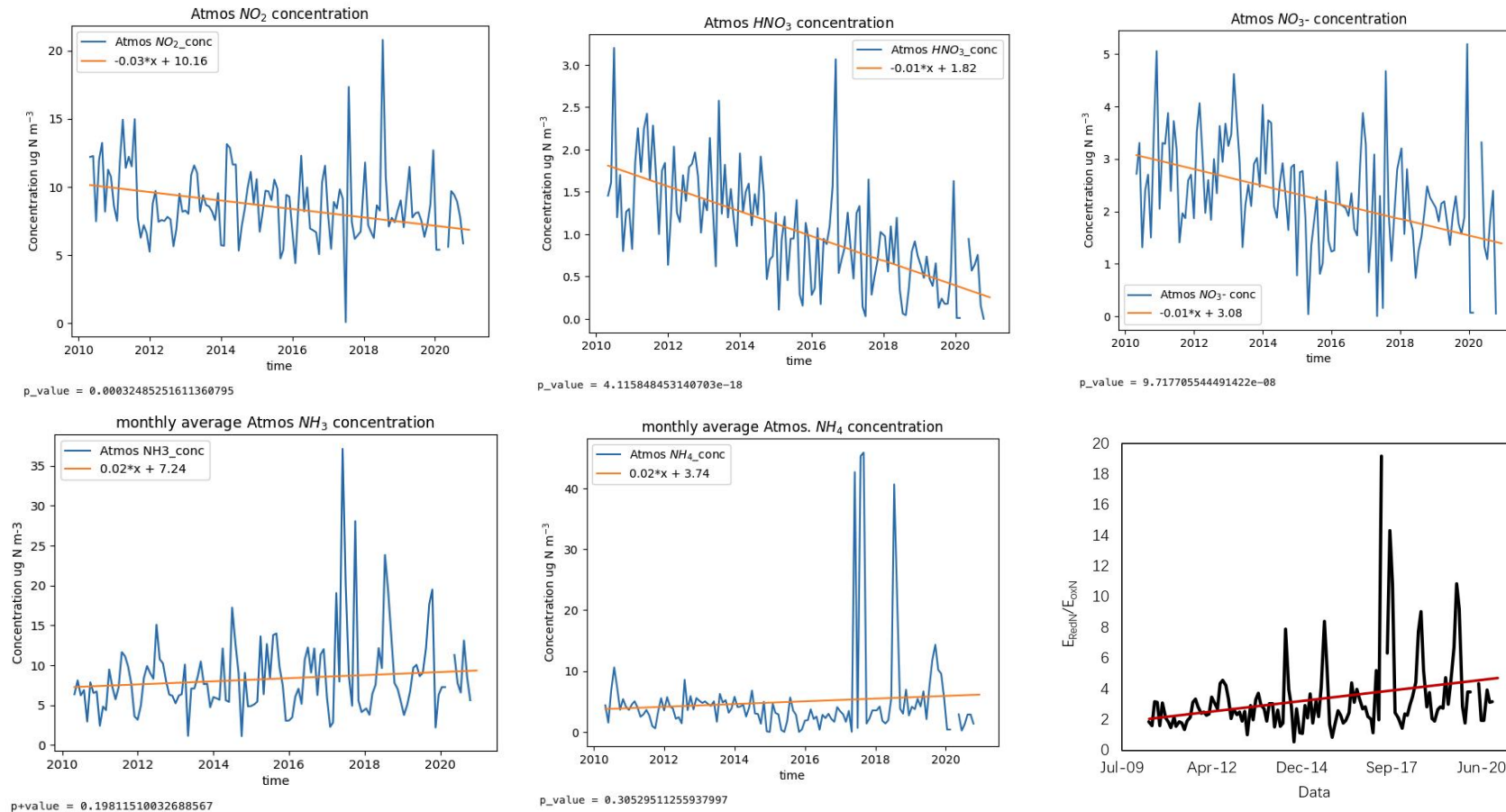


2007–2017年NH站点SNA与PM_{2.5}比值



2007–2017年NS站点SNA逐年变化

6.Preliminary Results 2.3 (continued) : Variation of Atmospheric N concentration at Baiyun station (2010-2020)

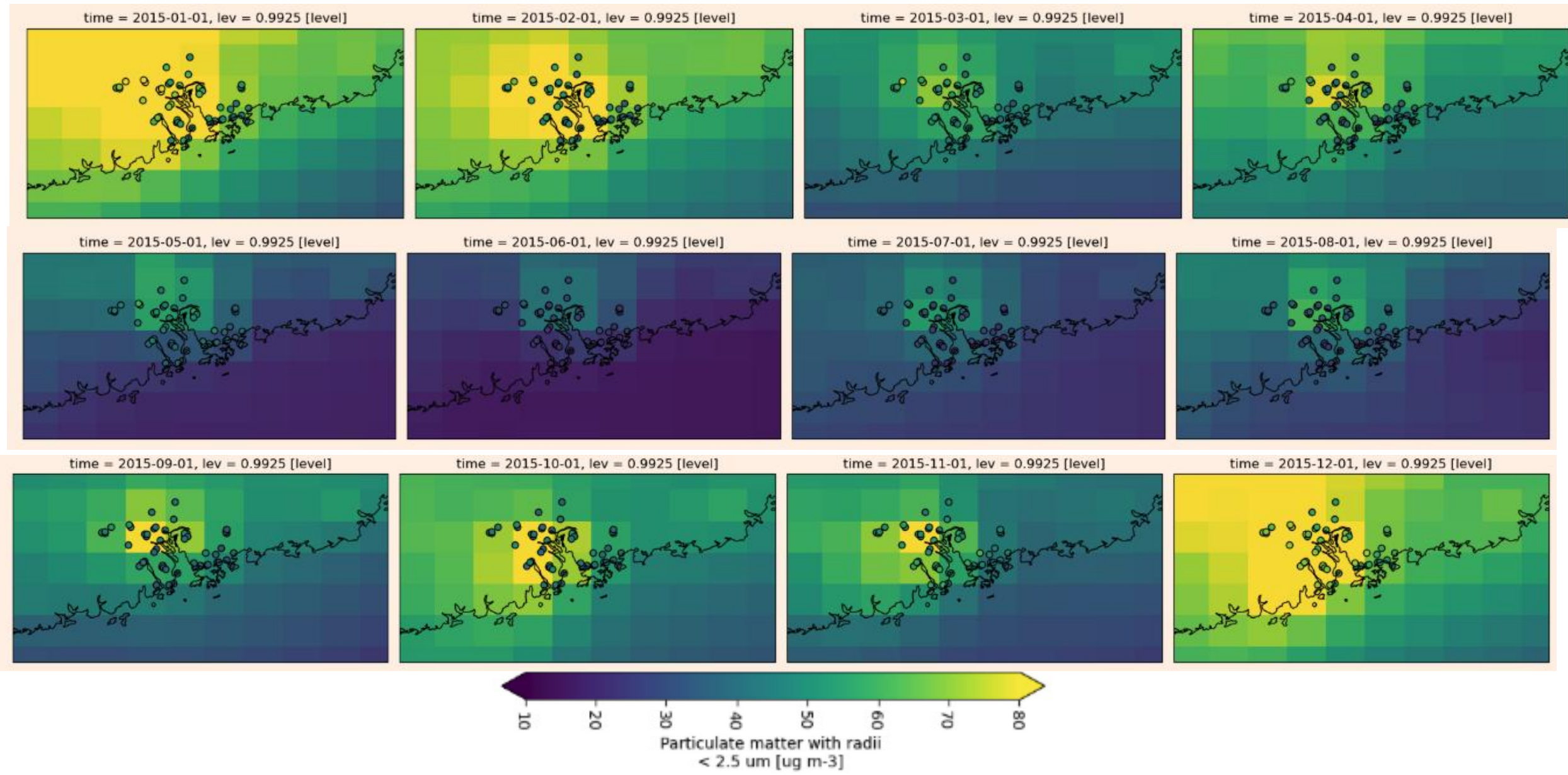


Define reduced /oxidized nitrogen
molar ratio:

$$R = \frac{\text{NH}_y}{\text{NO}_x}$$

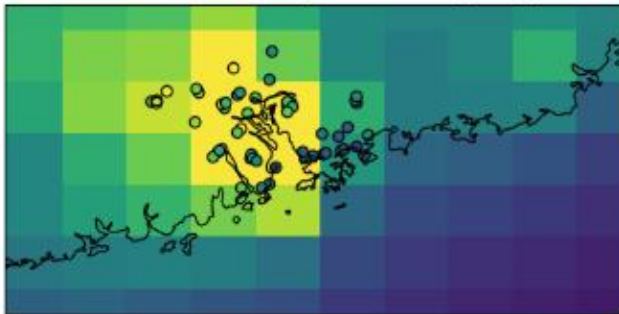
- Reduced nitrogen (NH_3 , NH_4^+) concentrations in the PRD continued to increase, while oxidized nitrogen (NO_x , NO_3^-) decreased gradually.
- The rising ratio of reduced-to-oxidized nitrogen may be fundamentally altering $\text{PM}_{2.5}$ composition.
- Diagnostic question: As SO_2 and NO_x continue to decline, has NH_3 become the critical bottleneck preventing further $\text{PM}_{2.5}$ reductions?

6. Preliminary Results 2.4: GEOS-Chem Simulated PM_{2.5} Component Changes

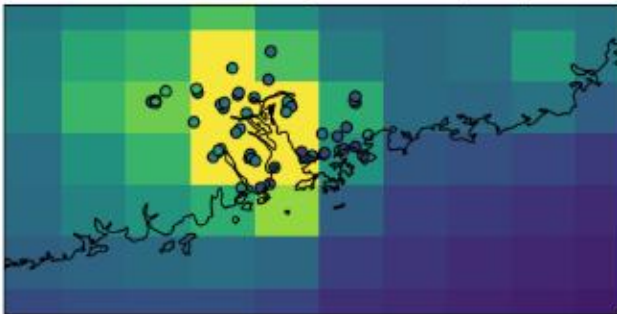


SO2

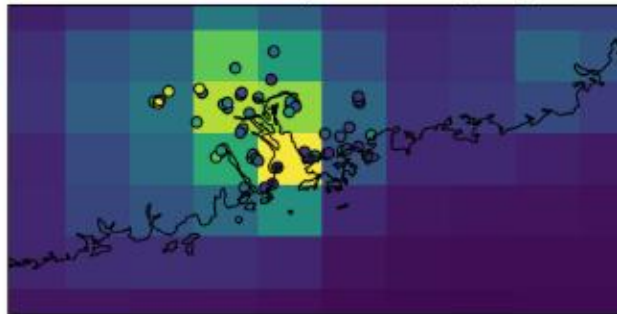
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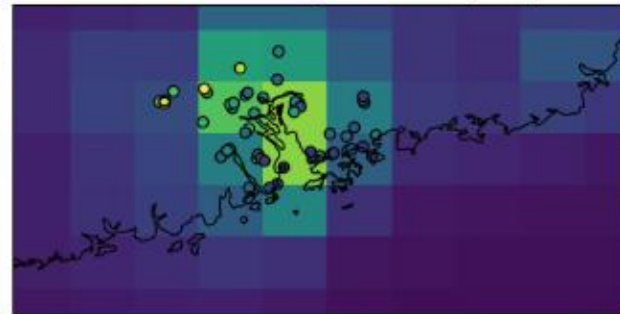
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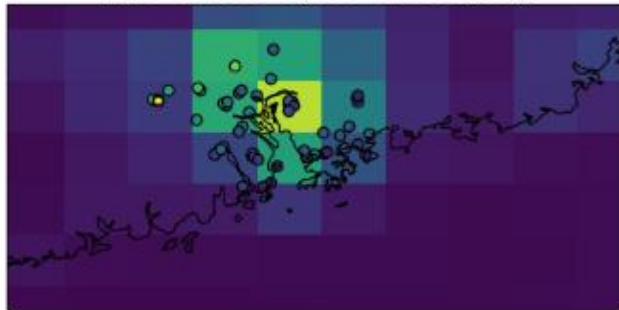
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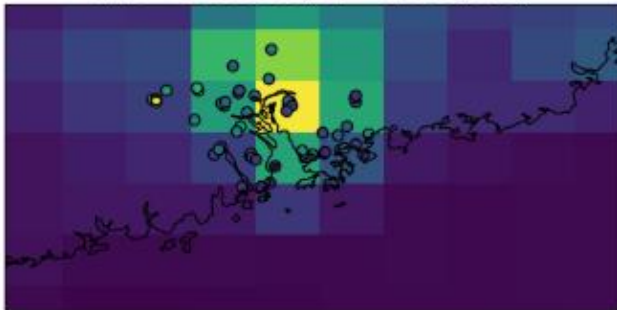
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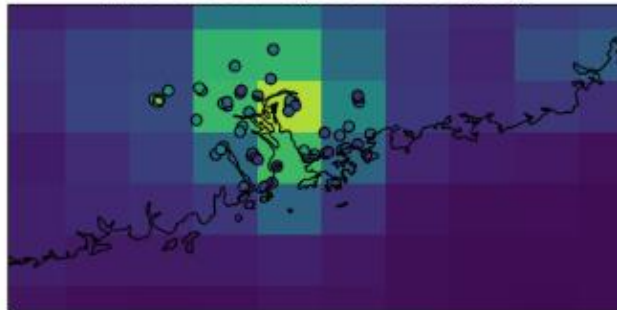
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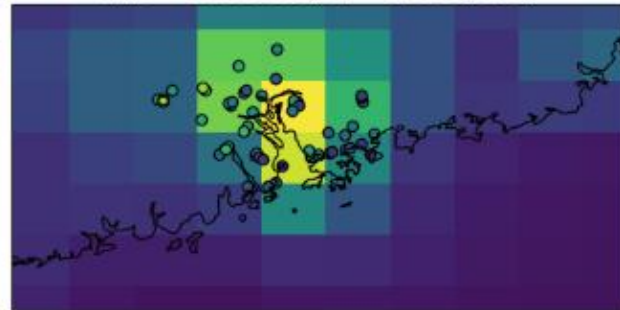
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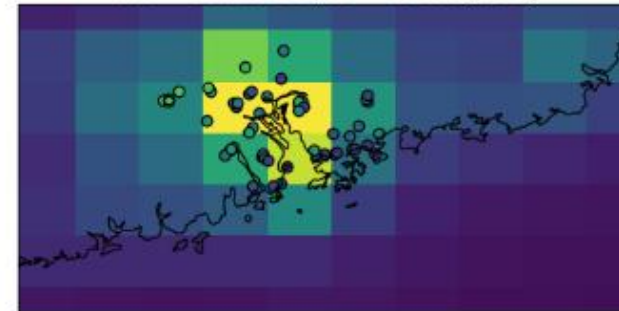
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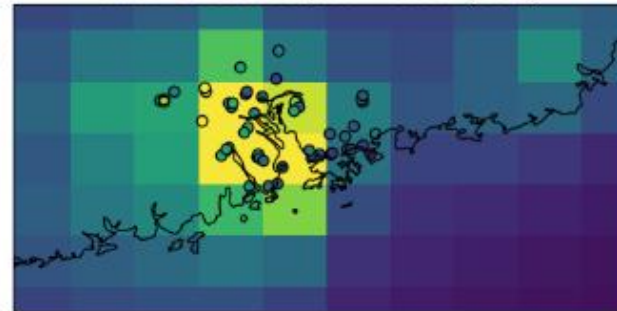
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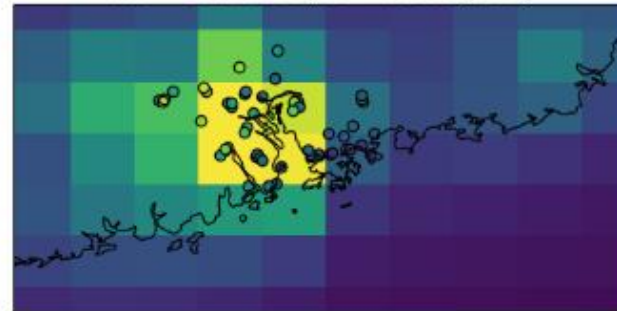
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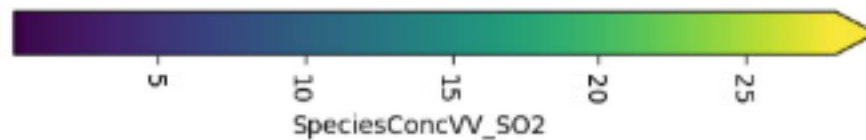
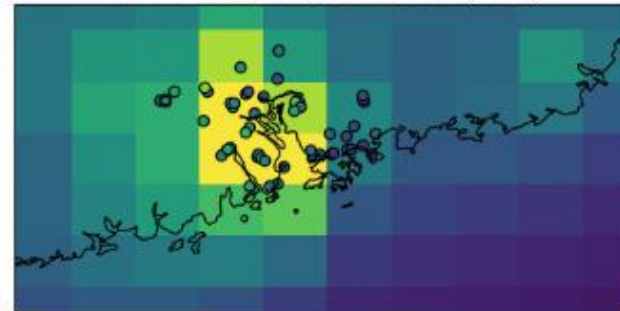
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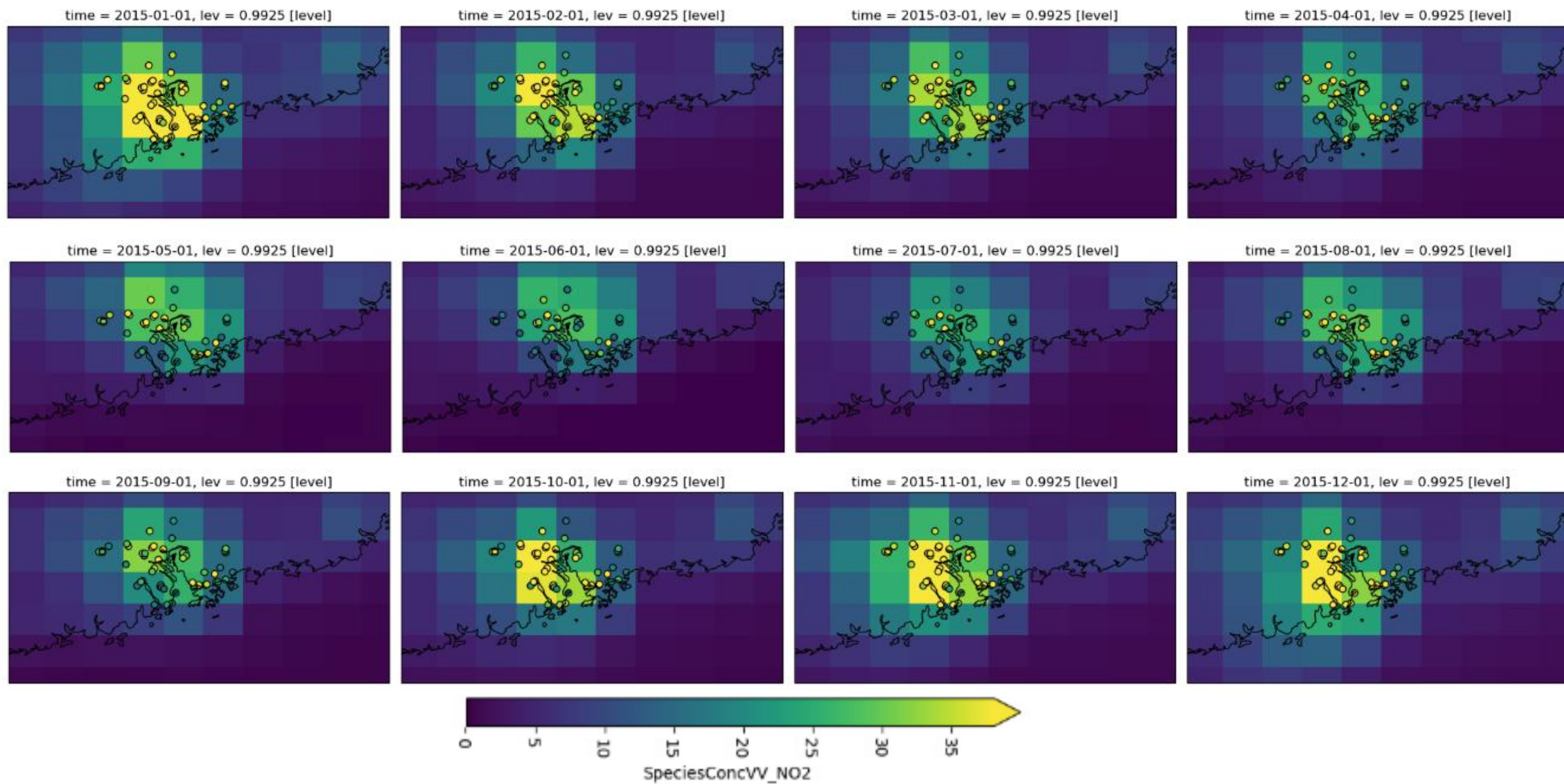
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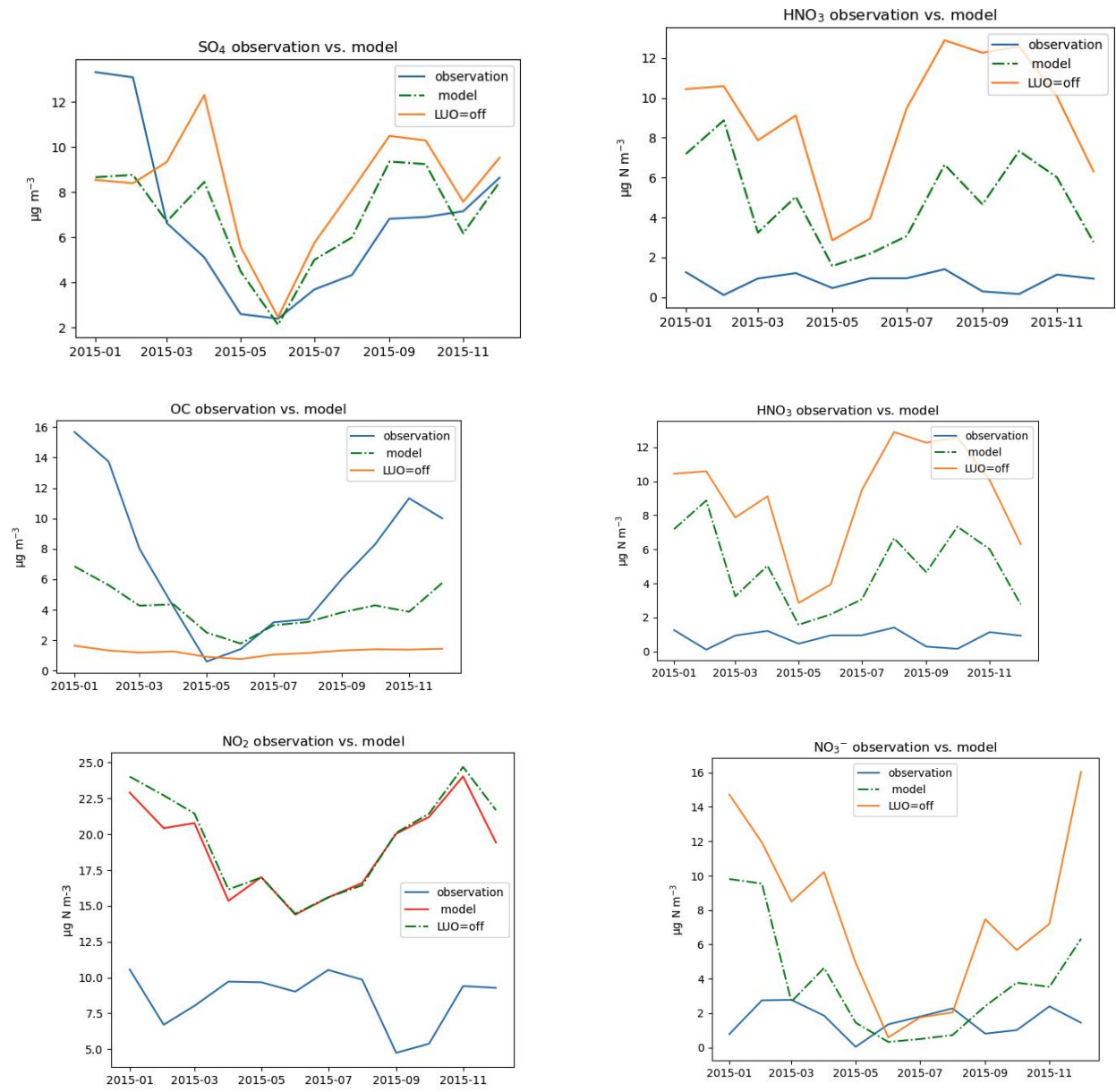
time = 2015-12-01, lev = 0.9925 [level]



NO2



其次比时间变化。若有**PM2.5**随时间变化，最好加上。



关于时间变化的对比，你参考画如下图：

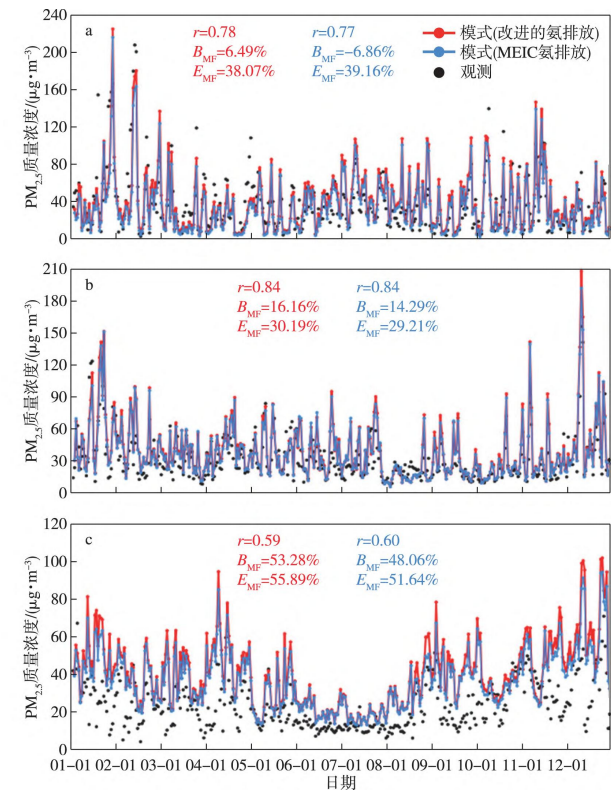
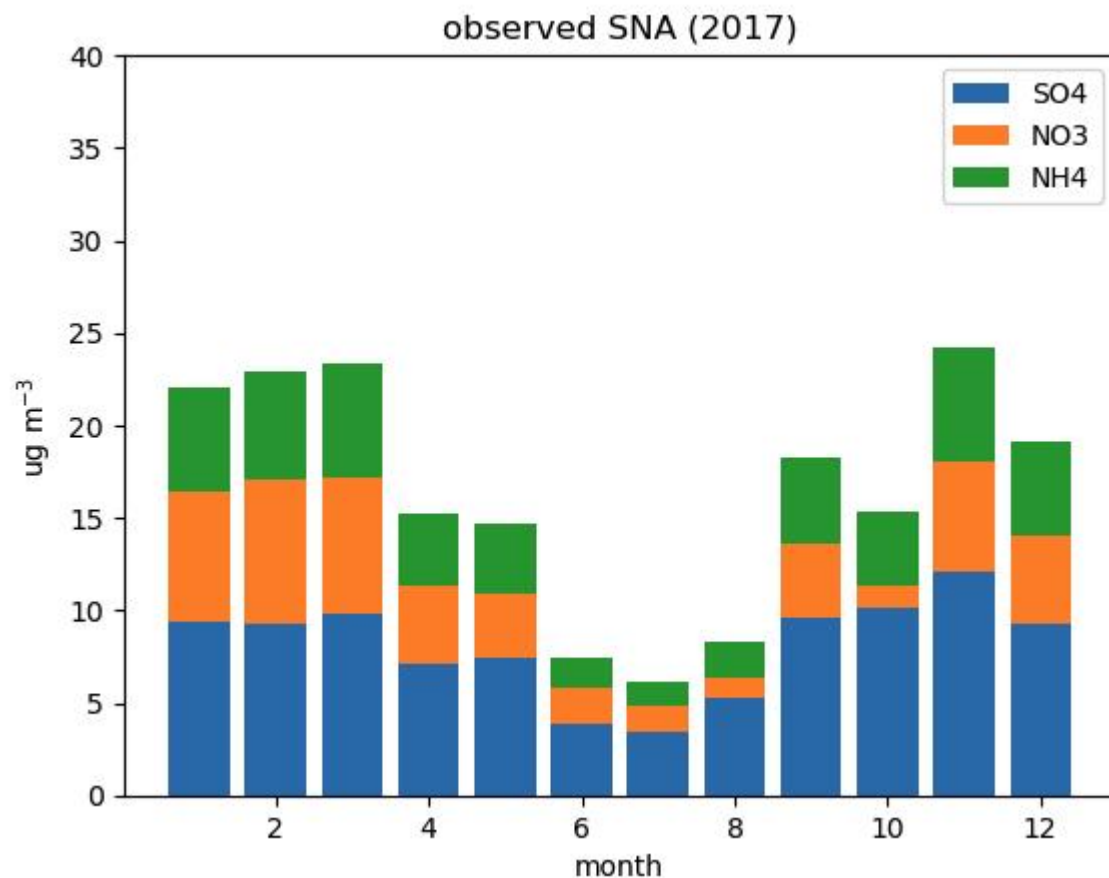
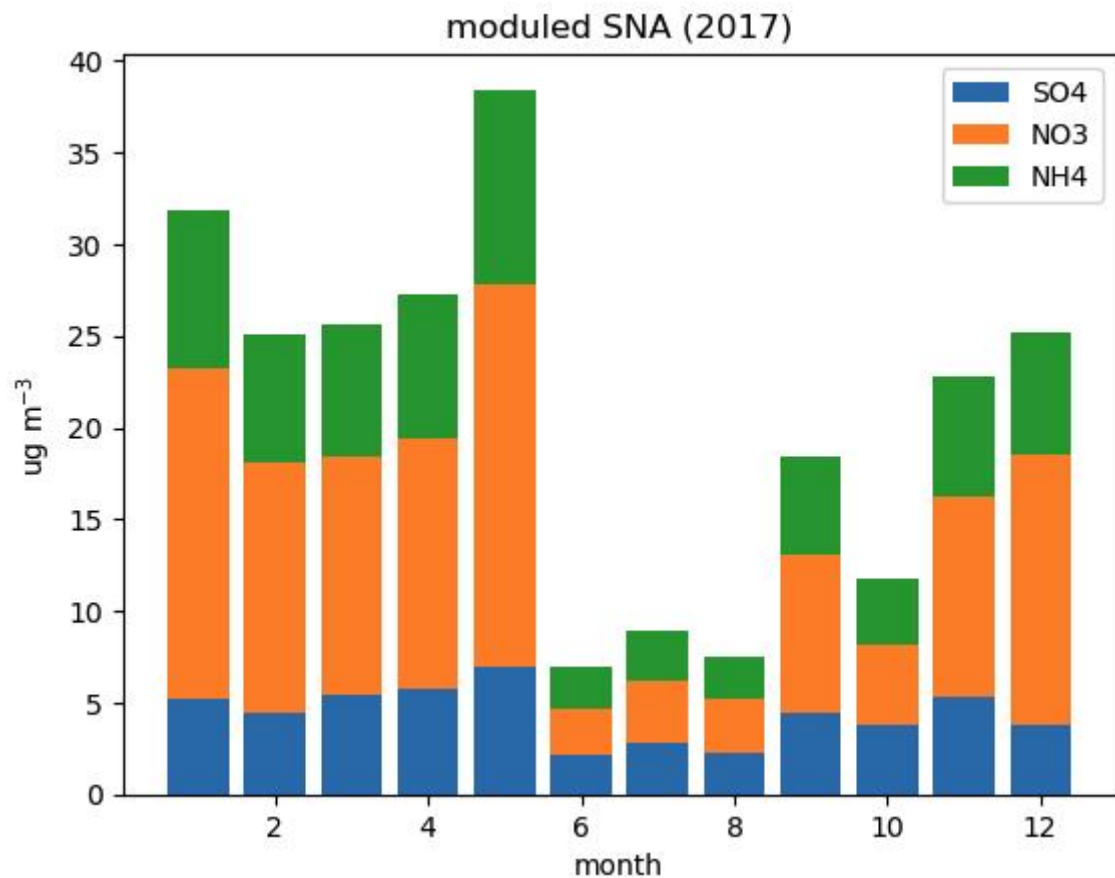


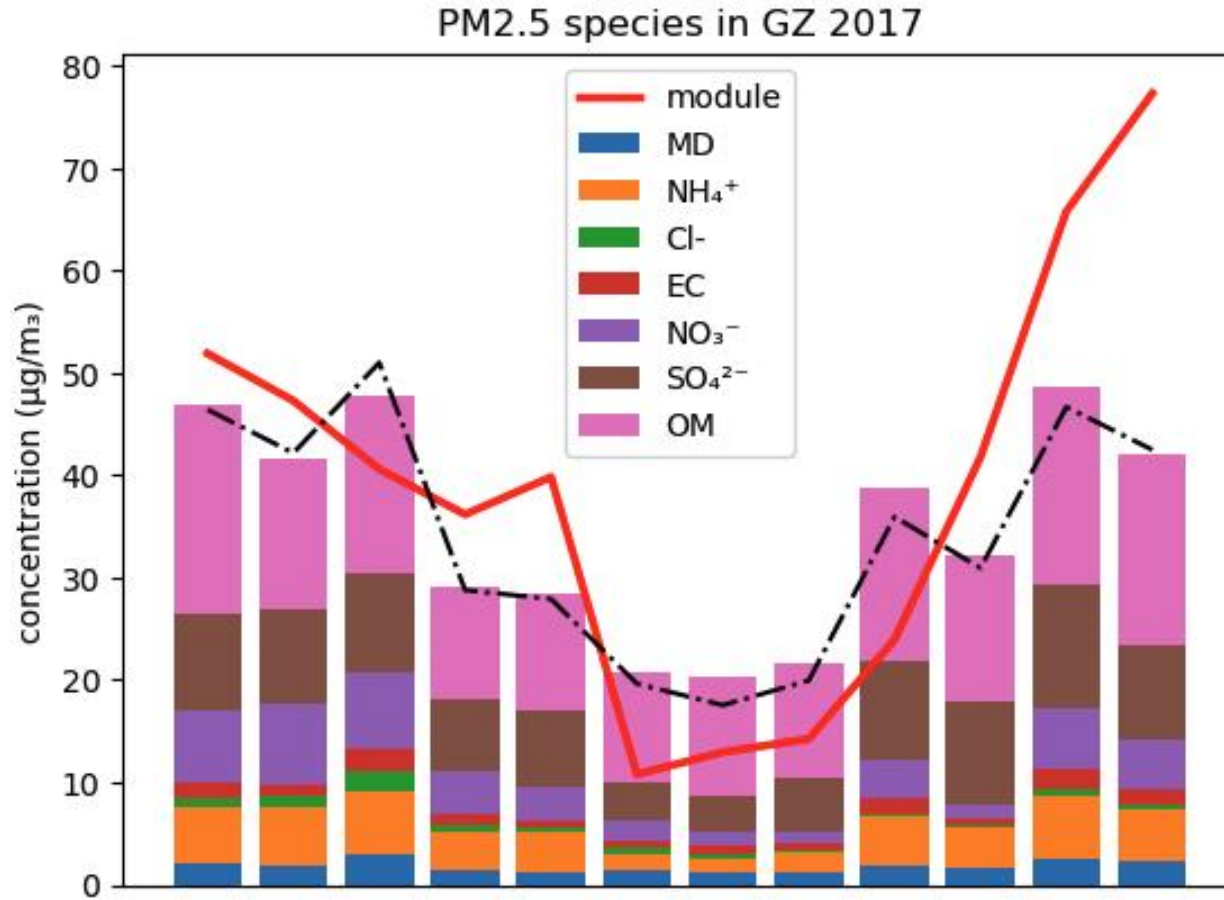
图 2 2020 年北京 (a)、上海 (b) 和 广州 (c) 地区模拟和观测的日均 PM 2.5 质量浓度 (单位: $\mu\text{g} \cdot \text{m}^{-3}$; 蓝色折线为原始的 MEIC 氨排放模拟结果, 红色折线为采用同化改善氨排放后的模拟结果)

第三比组分变化



- observed summer SNA is only 1/3 of winter SNA
- module summer SNA is only about 1/5 of winter SNA
- 如果能解释一下原因更好
- 探讨不同季节 SO_4 , NO_3 和 NH_4 变化特征

下图是你模拟的？还是观测值，如是观测的，你可把你的模拟值画类似的图。图中横轴没有单位和说明。



7. Future Work & Expected Outcomes

Key Questions:

1. Has sulfate decline altered ammonia partitioning, thereby enhancing nitrate formation potential?
2. How does the oxidized-to-reduced nitrogen emission structure affect secondary aerosol formation, $\text{PM}_{2.5}$, and O_3 ?
3. What are the atmospheric nitrogen deposition fluxes and their ecological consequences in the PRD?
4. What future emission-reduction scenarios and $\text{PM}_{2.5}$ – O_3 synergistic control pathways are optimal for the PRD?

Expected Outcomes:

- A closed-loop analysis from emissions → chemical transformation → deposition → policy scenario for PRD reactive nitrogen.
- A science-based, region-specific roadmap for transitioning PRD air quality management into the “ NH_3 -sensitive” era.

Thanks a lot!